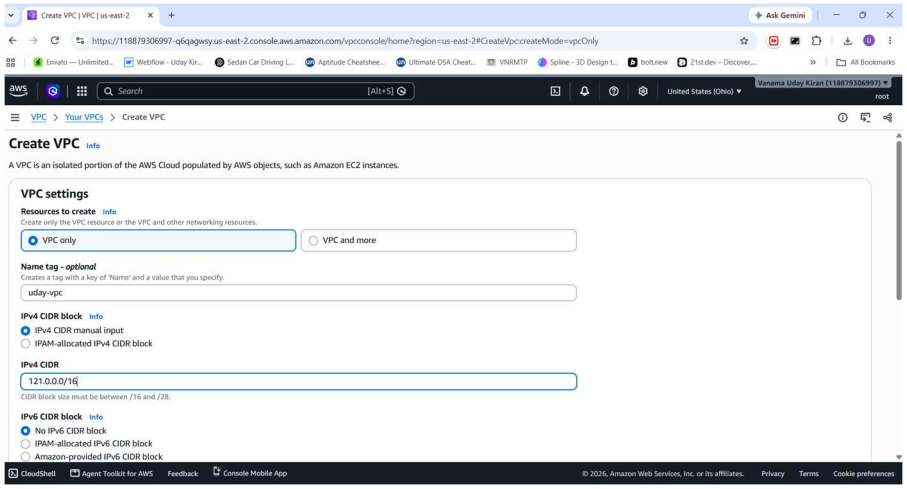
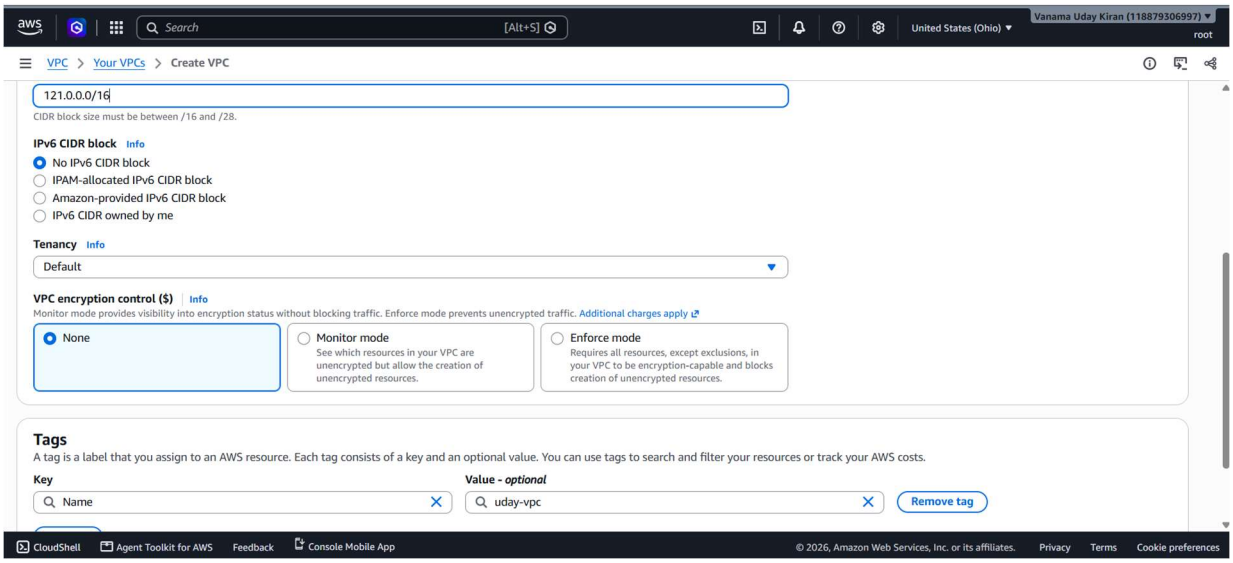
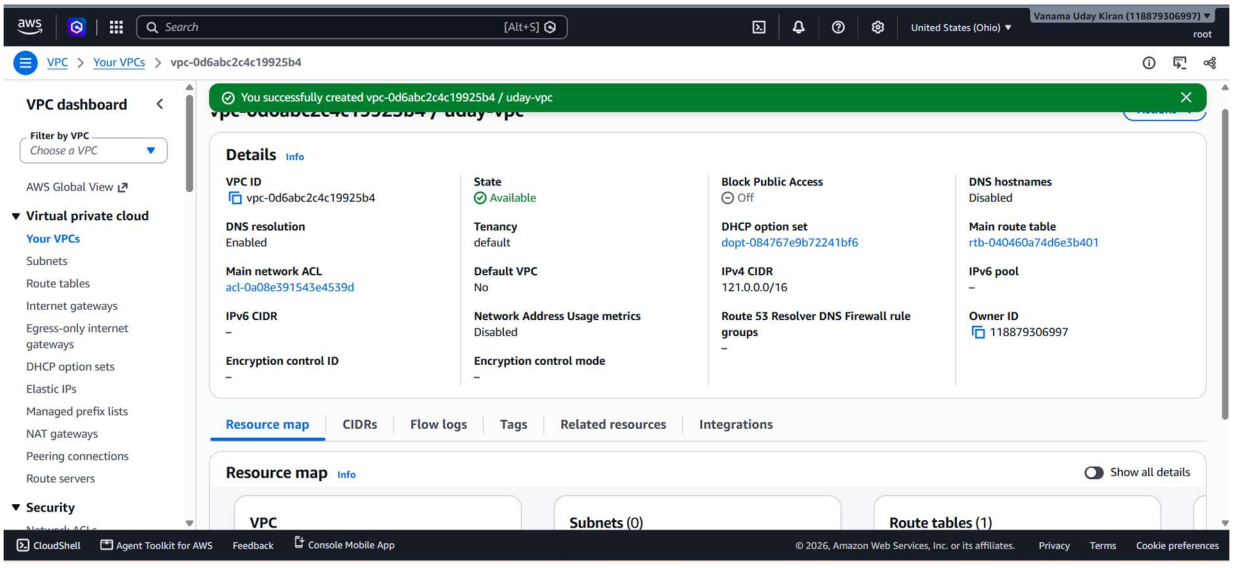
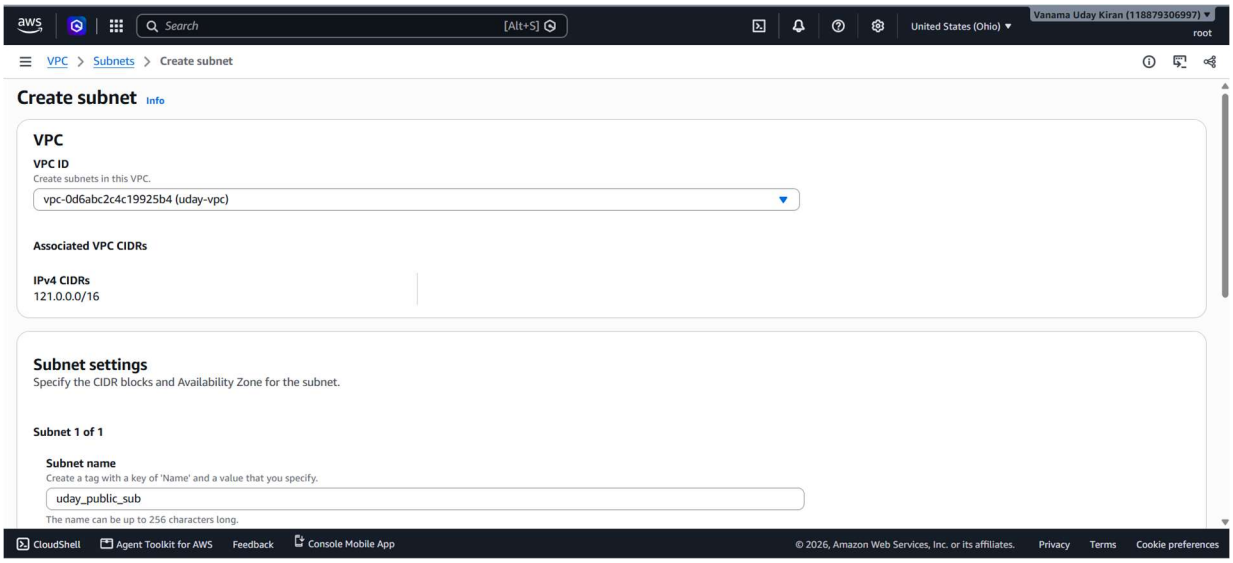
Creating the VPC With the CIDR 121.0.0.0/16

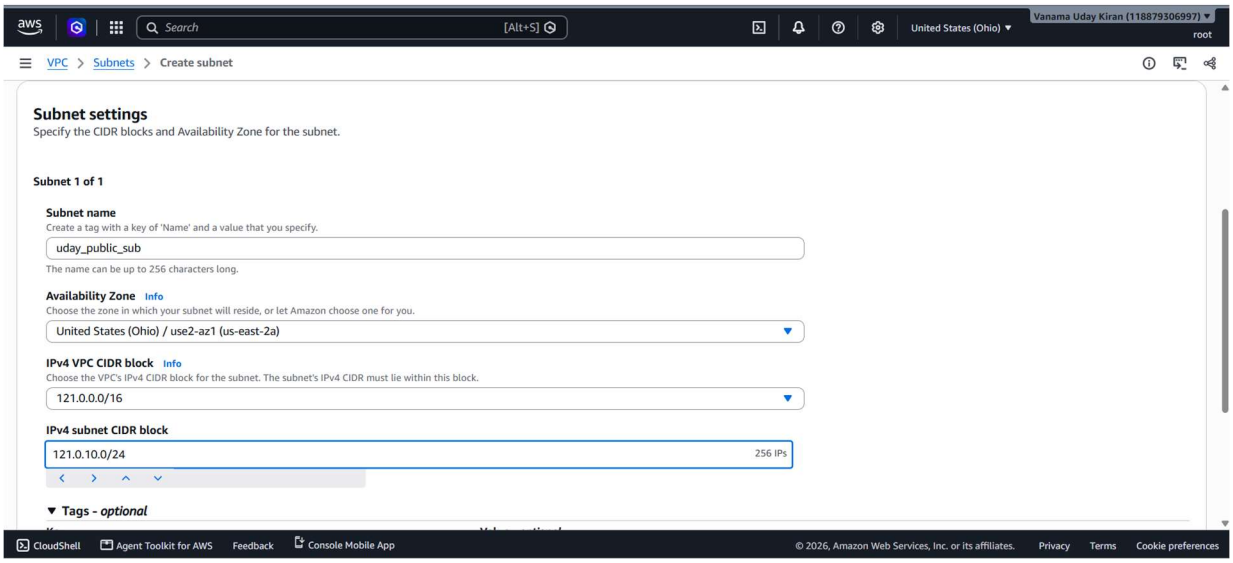


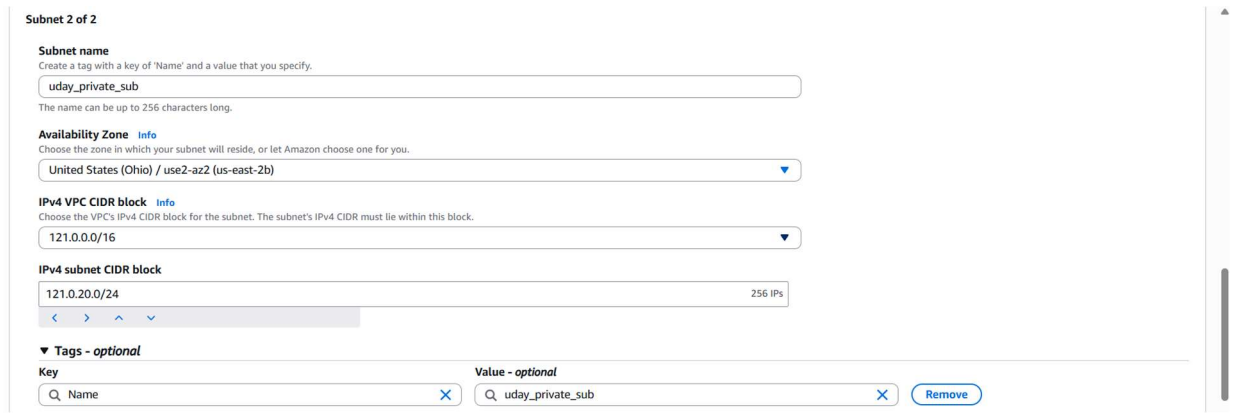


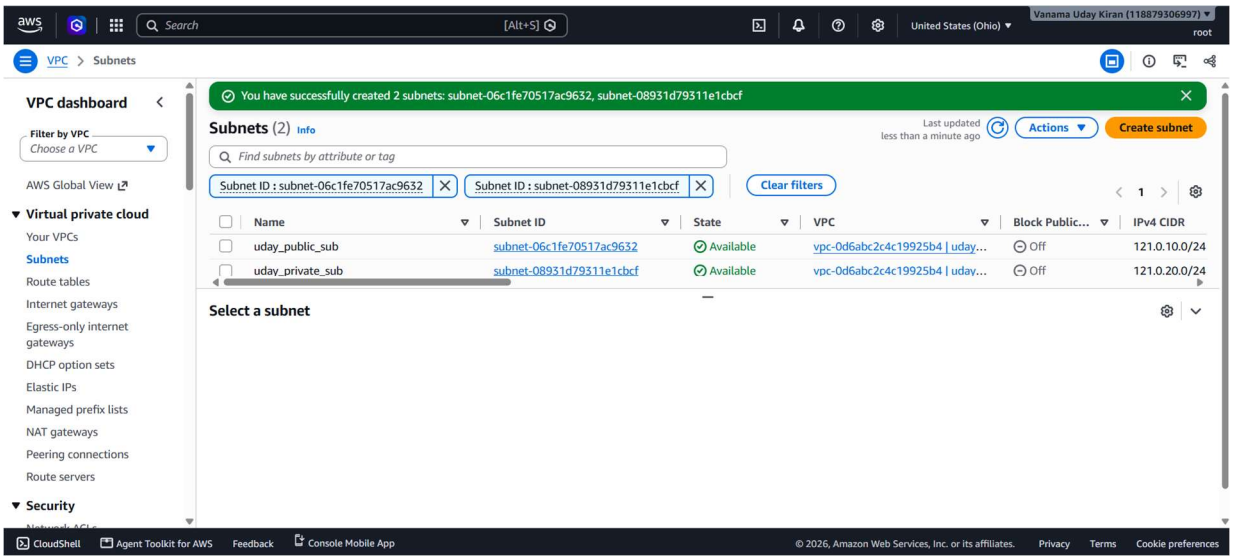


Creating the Subnets :

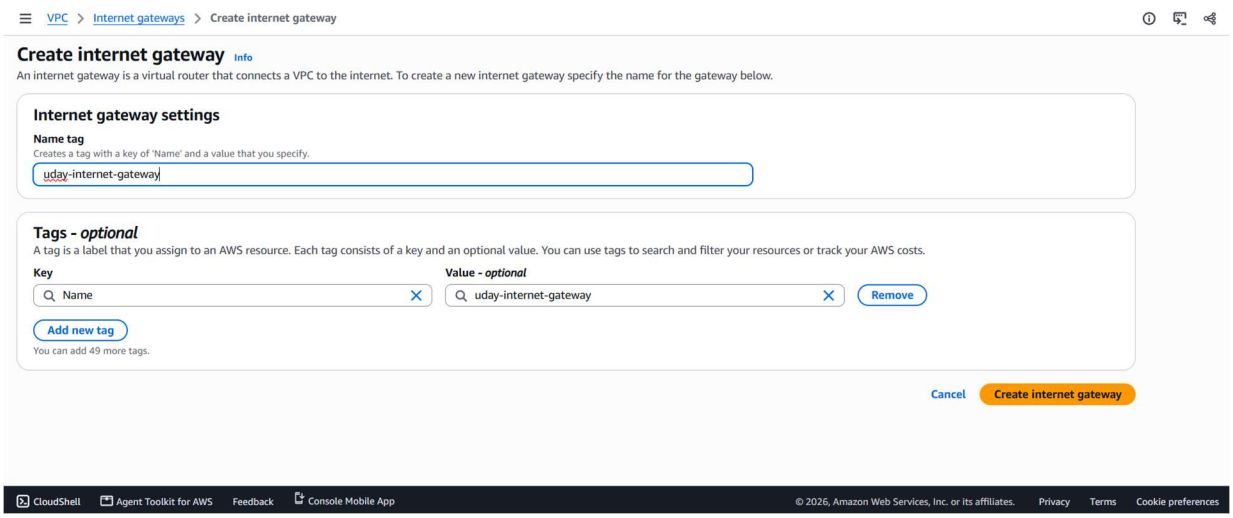




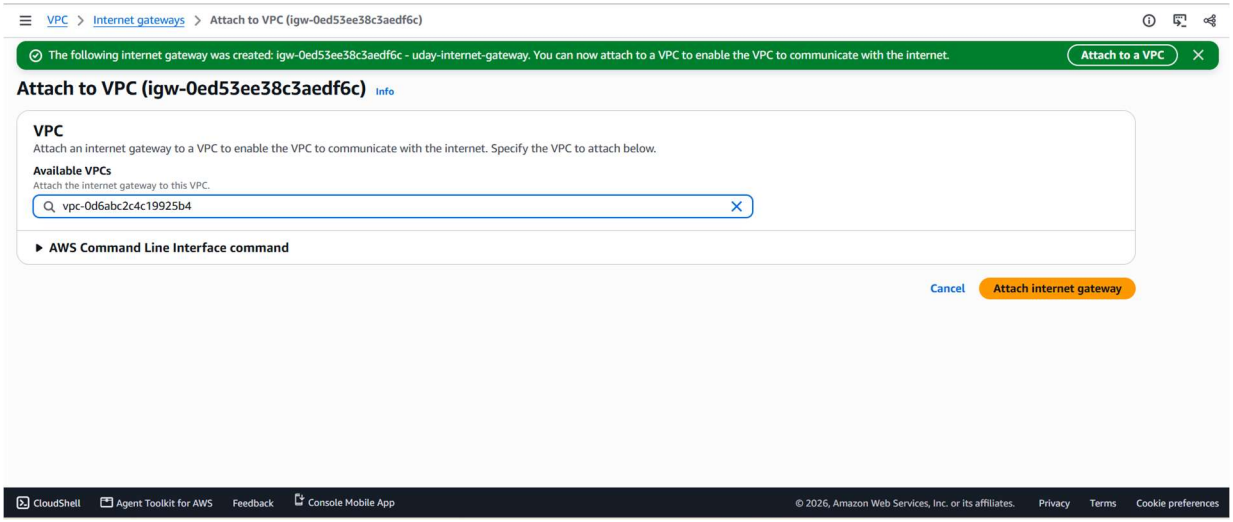


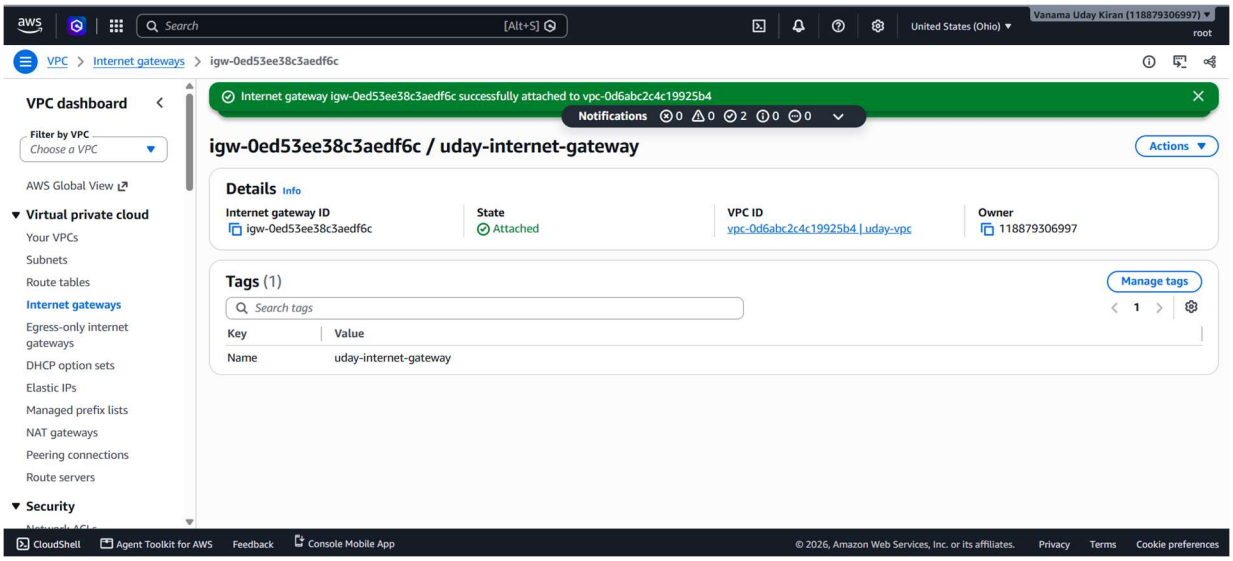


Creating the Internet gateway:

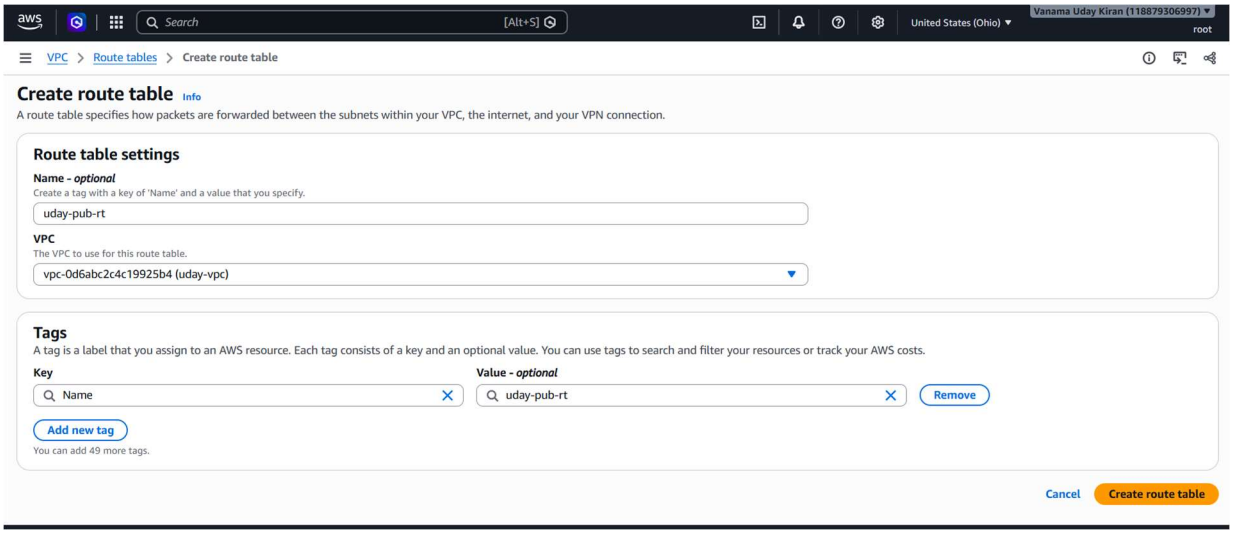


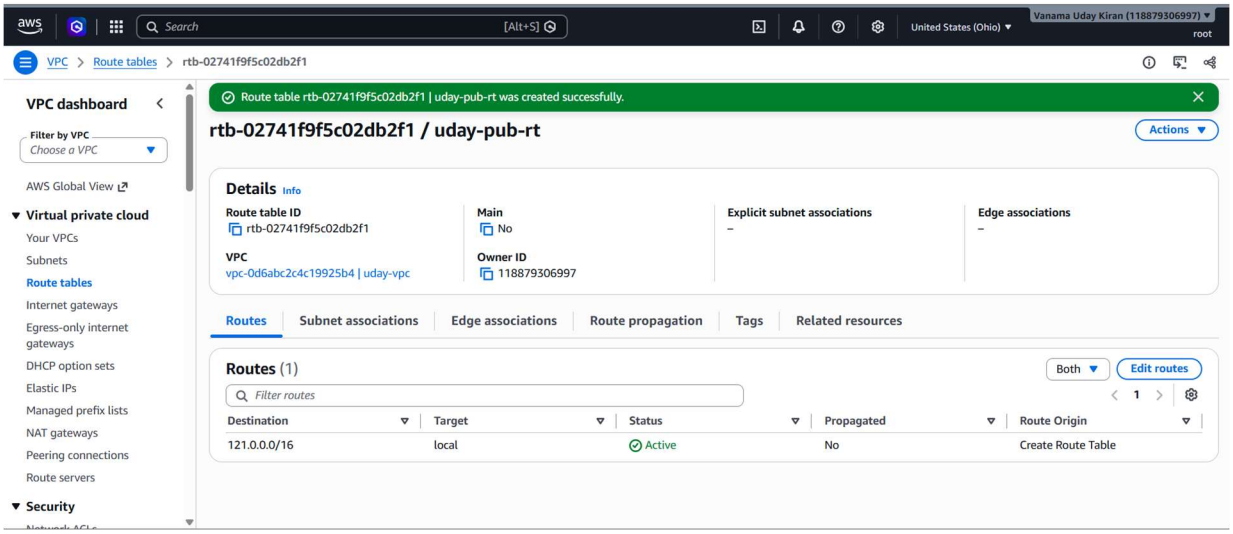
Attach it to VPC :

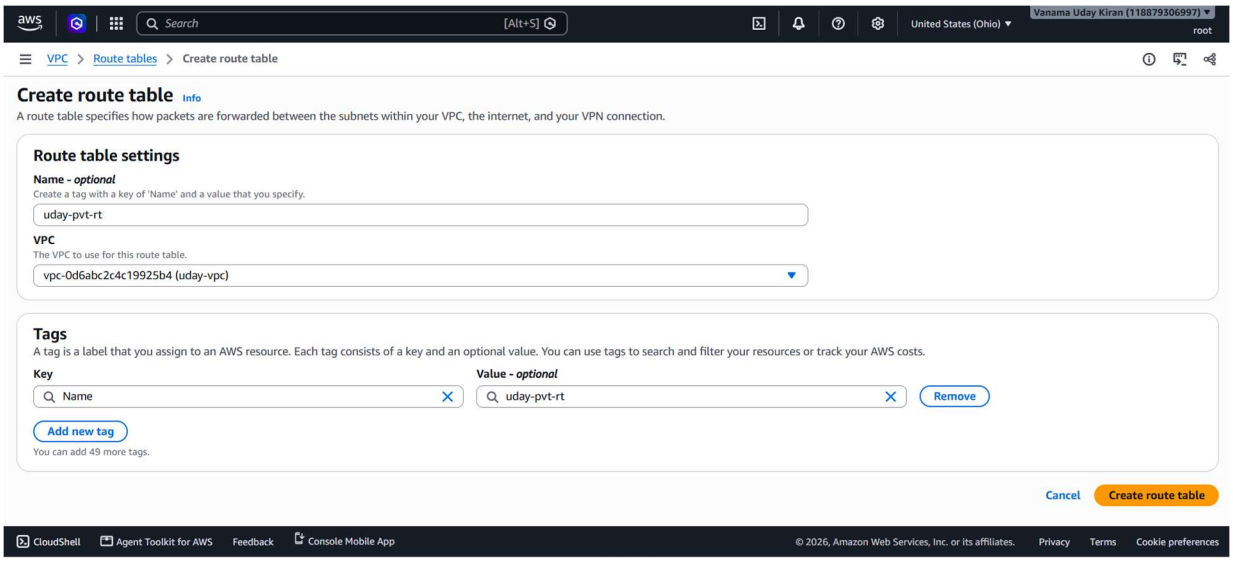


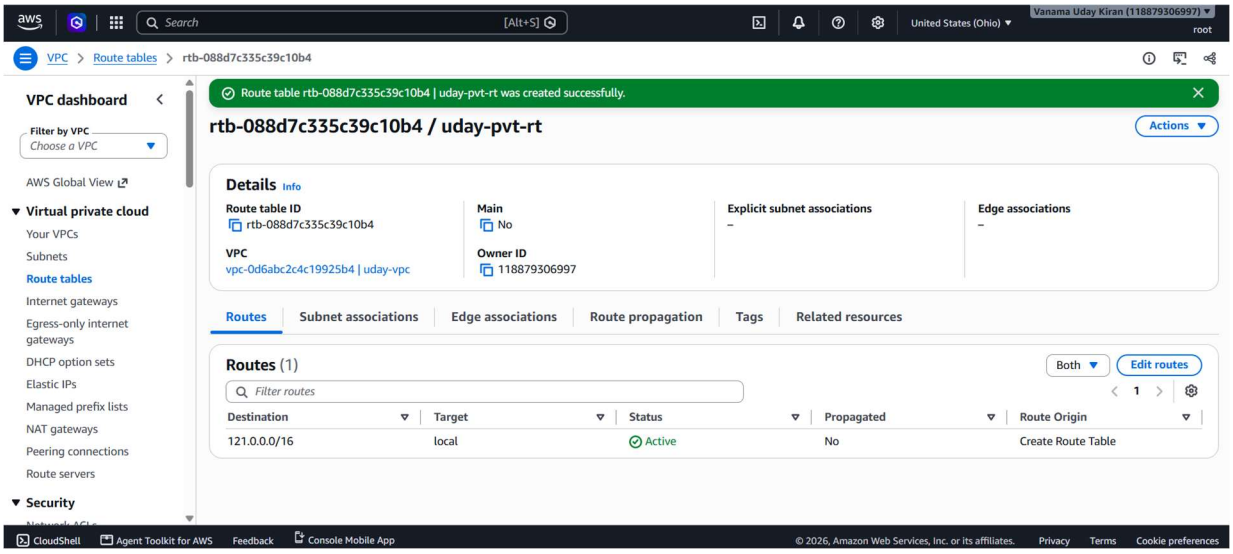


Creating The Route Tables:

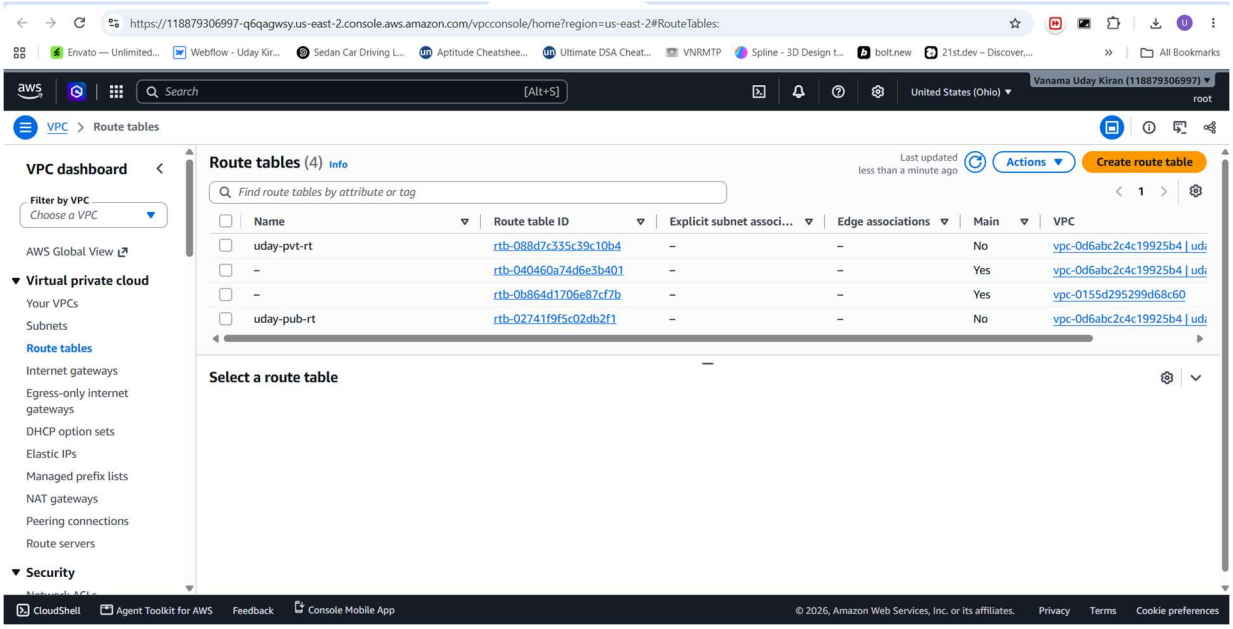





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root
VPC > Route tables > Create route table
Create route table Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
uday-pvt-rt
VPC
The VPC to use for this route table.
vpc-0d6abc2c4c19925b4 (uday-vpc)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
uday-pvt-rt
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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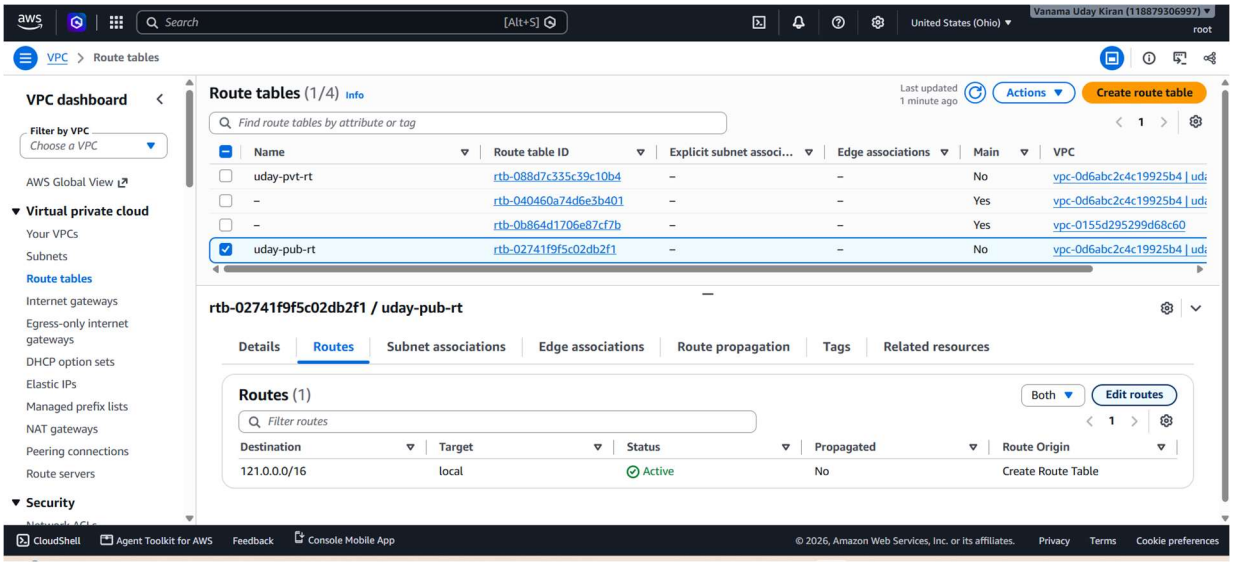

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root
VPC > Route tables > rtb-088d7c335c39c10b4
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Route table rtb-088d7c335c39c10b4 | uday-pvt-rt was created successfully.
rtb-088d7c335c39c10b4 / uday-pvt-rt
Actions
Details Info
Route table ID
rtb-088d7c335c39c10b4
Main
No
Explicit subnet associations
–
Edge associations
–
VPC
vpc-0d6abc2c4c19925b4 | uday-vpc
Owner ID
118879306997
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (1)
Both
Edit routes
Filter routes
1
Destination
Target
Status
Propagated
Route Origin
121.0.0.0/16
local
Active
No
Create Route Table
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Checking :

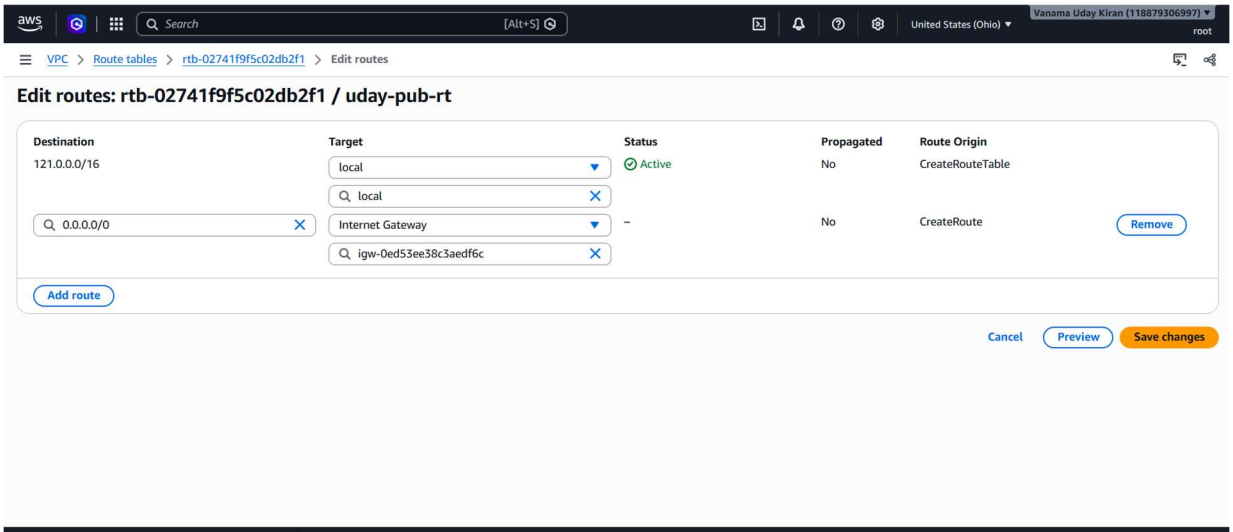


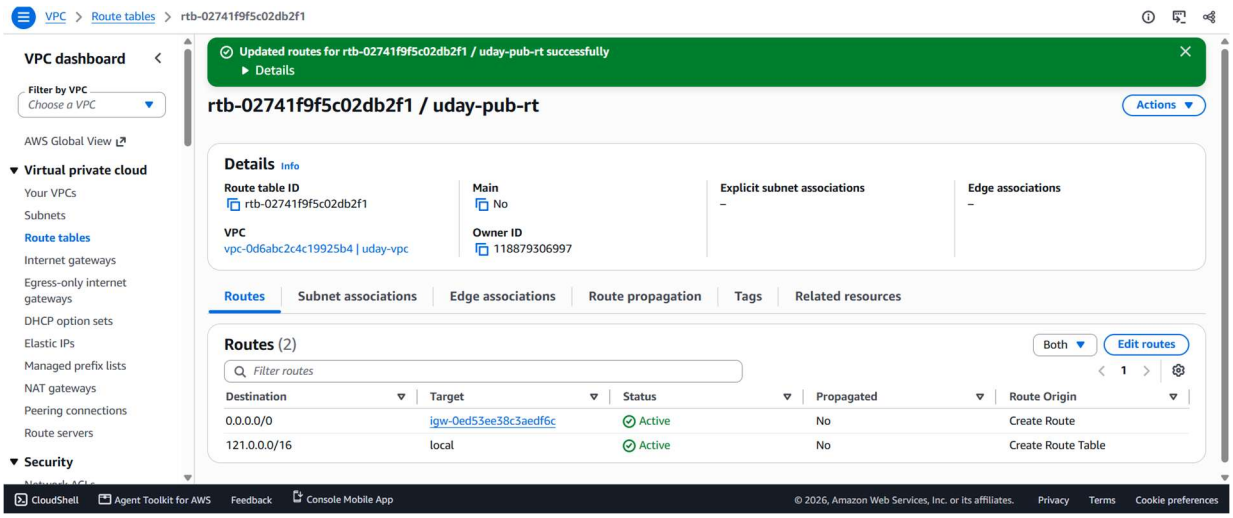
Configuring the Public Route Tables:

Connecting the Internet gateway to the Public rt:

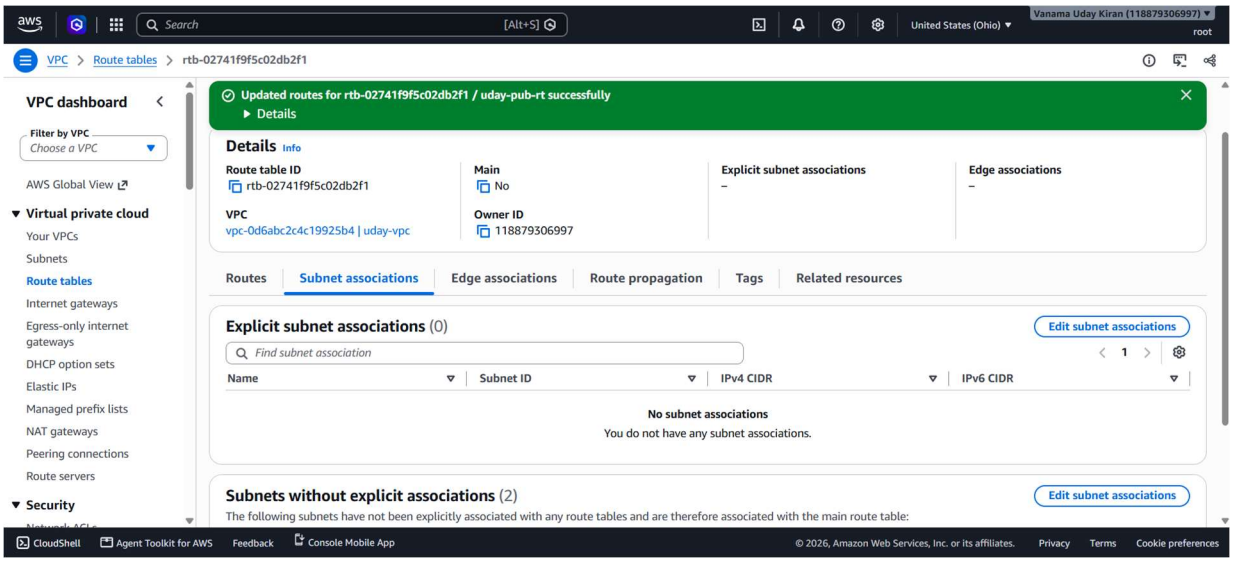


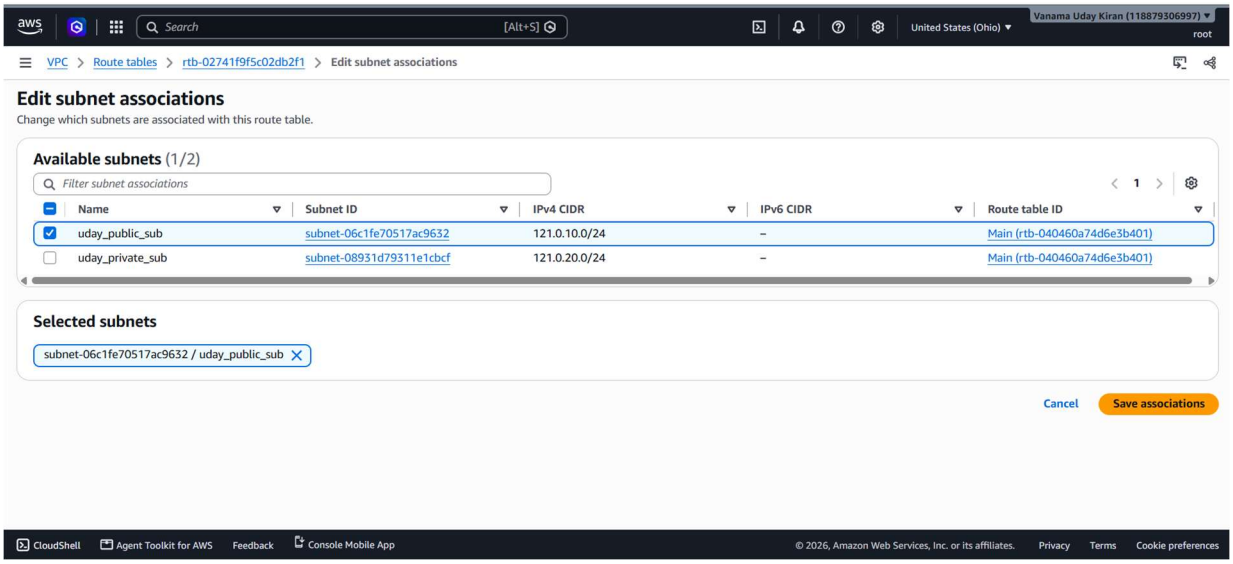
Edit the Rules:

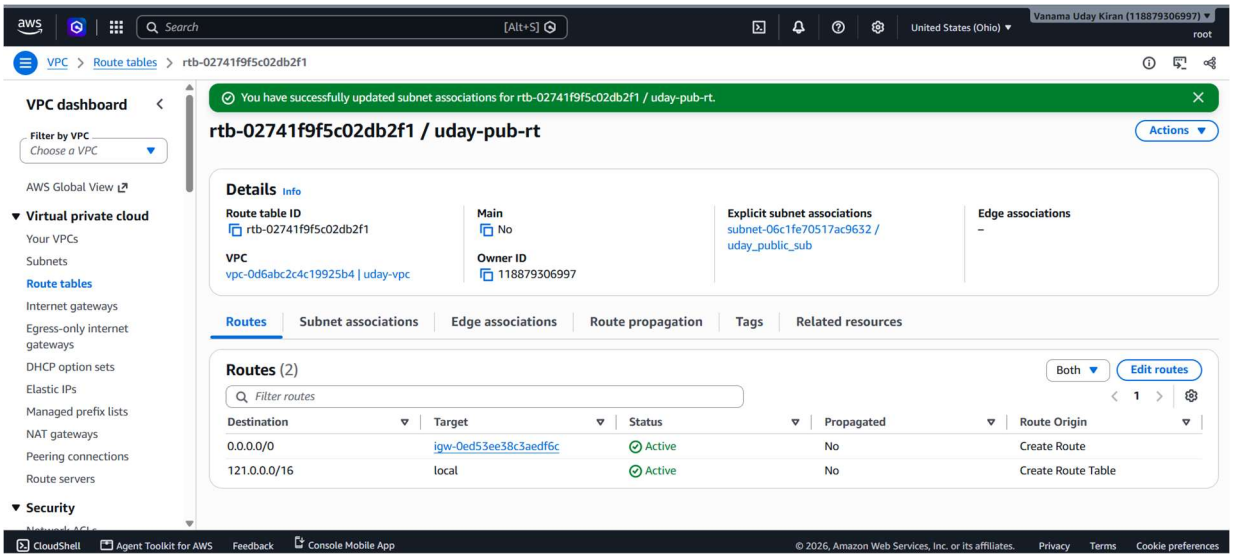




Connecting the Pub_rt to the Pub Subnet :

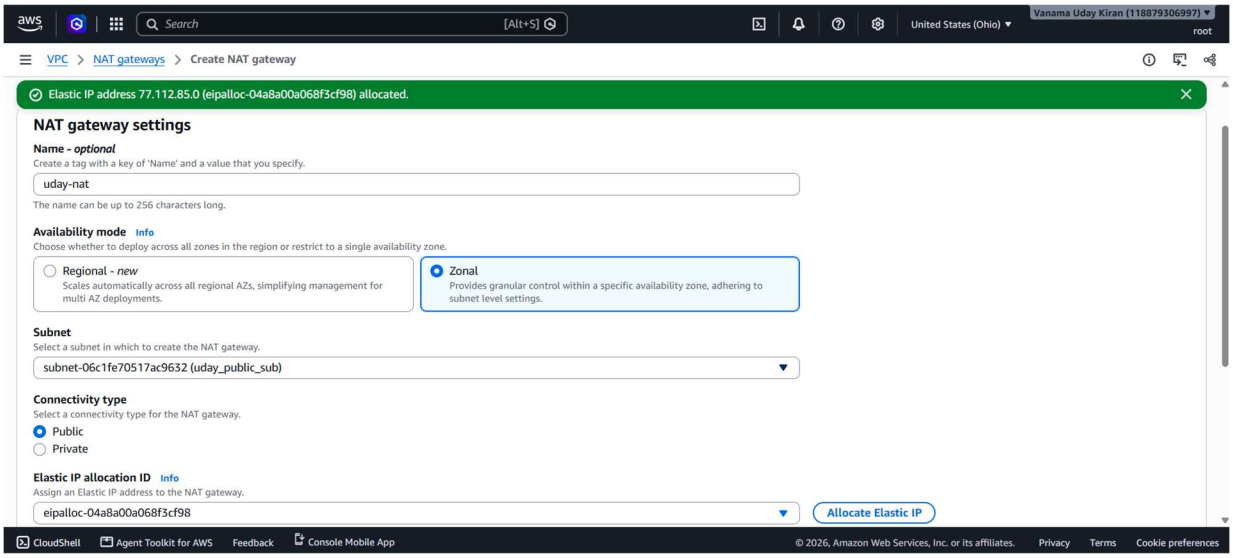

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United States (Ohio)
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root
VPC > Route tables > rtb-02741f9f5c02db2f1
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Updated routes for rtb-02741f9f5c02db2f1 / uday-pub-rt successfully
Details
Details Info
Route table ID
rtb-02741f9f5c02db2f1
Main
No
Explicit subnet associations
–
Edge associations
–
VPC
vpc-0d6abc2c4c19925b4 | uday-vpc
Owner ID
118879306997
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Explicit subnet associations (0)
Edit subnet associations
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
No subnet associations
You do not have any subnet associations.
Subnets without explicit associations (2)
Edit subnet associations
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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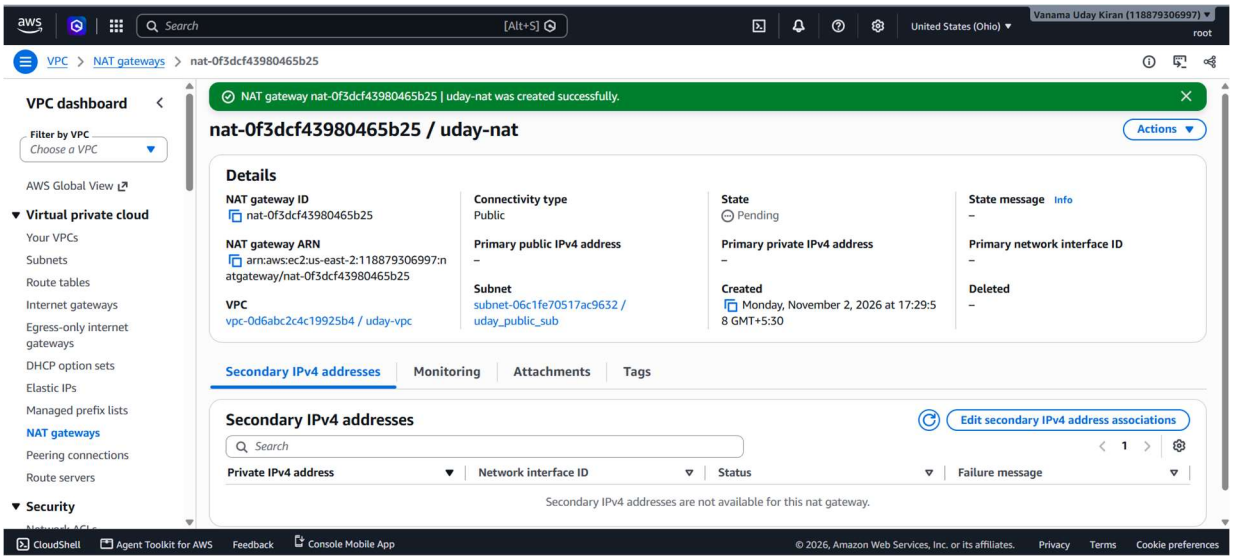

Search [Alt+S]
United States (Ohio)
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root
VPC > Route tables > rtb-02741f9f5c02db2f1 > Edit subnet associations
Edit subnet associations
Change which subnets are associated with this route table.
Available subnets (1/2)
Filter subnet associations
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
Route table ID
uday_public_sub
subnet-06c1fe70517ac9632
121.0.10.0/24
–
Main (rtb-040460a74d6e3b401)
uday_private_sub
subnet-08931d79311e1cbcf
121.0.20.0/24
–
Main (rtb-040460a74d6e3b401)
Selected subnets
subnet-06c1fe70517ac9632 / uday_public_sub
Cancel
Save associations
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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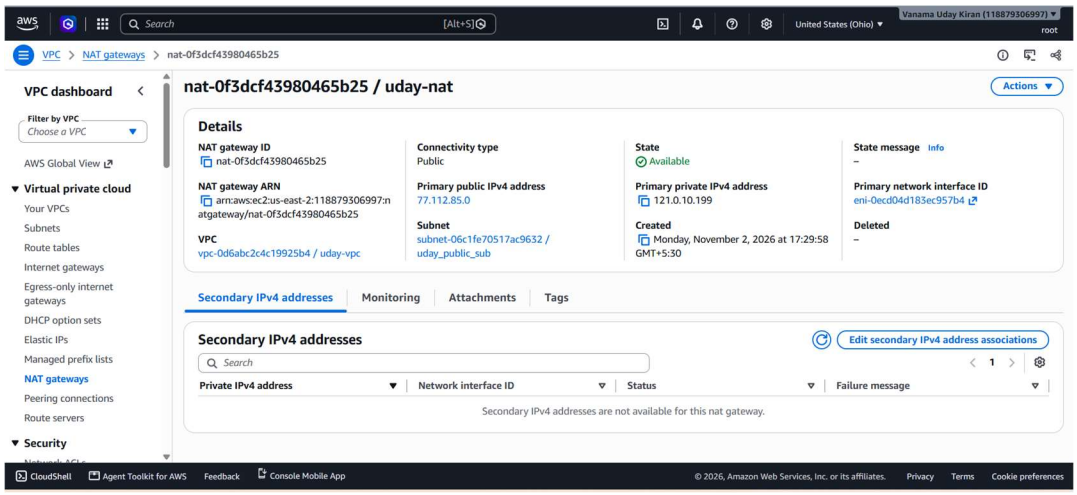

Search [Alt+S]
United States (Ohio)
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root
VPC > Route tables > rtb-02741f9f5c02db2f1
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
You have successfully updated subnet associations for rtb-02741f9f5c02db2f1 / uday-pub-rt.
rtb-02741f9f5c02db2f1 / uday-pub-rt
Actions
Details Info
Route table ID
rtb-02741f9f5c02db2f1
Main
No
Explicit subnet associations
subnet-06c1fe70517ac9632 / uday_public_sub
Edge associations
–
VPC
vpc-0d6abc2c4c19925b4 | uday-vpc
Owner ID
118879306997
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
igw-0ed53ee38c3aedf6c
Active
No
Create Route
121.0.0.0/16
local
Active
No
Create Route Table
CloudShell
Agent Toolkit for AWS
Feedback
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Configuring the Private Route Tables:

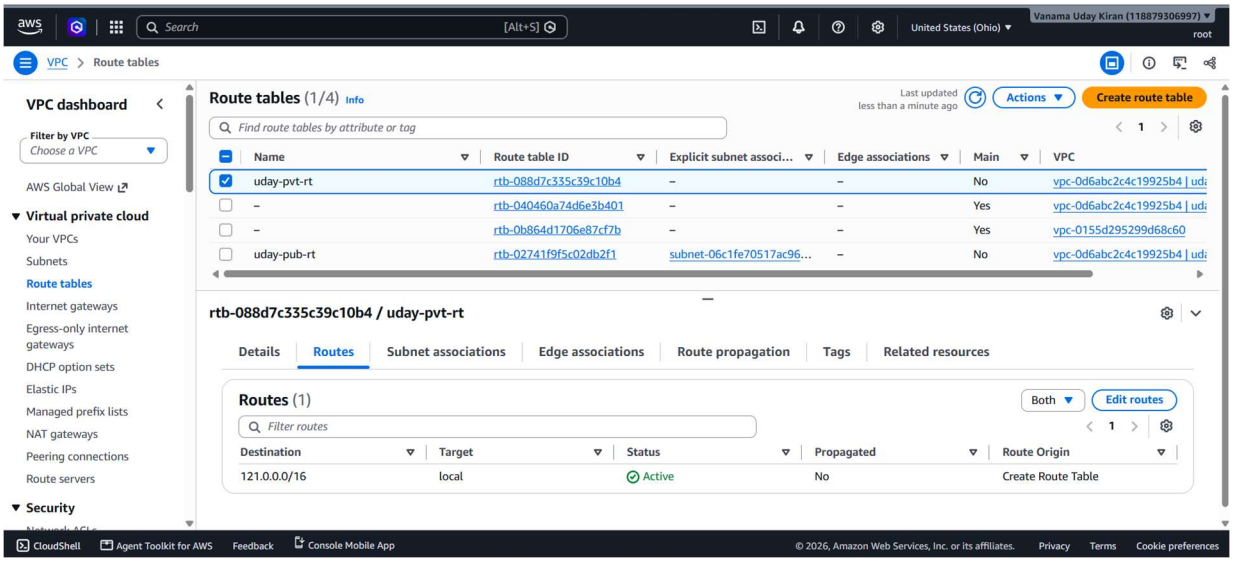
Creating The NAT Gateway :

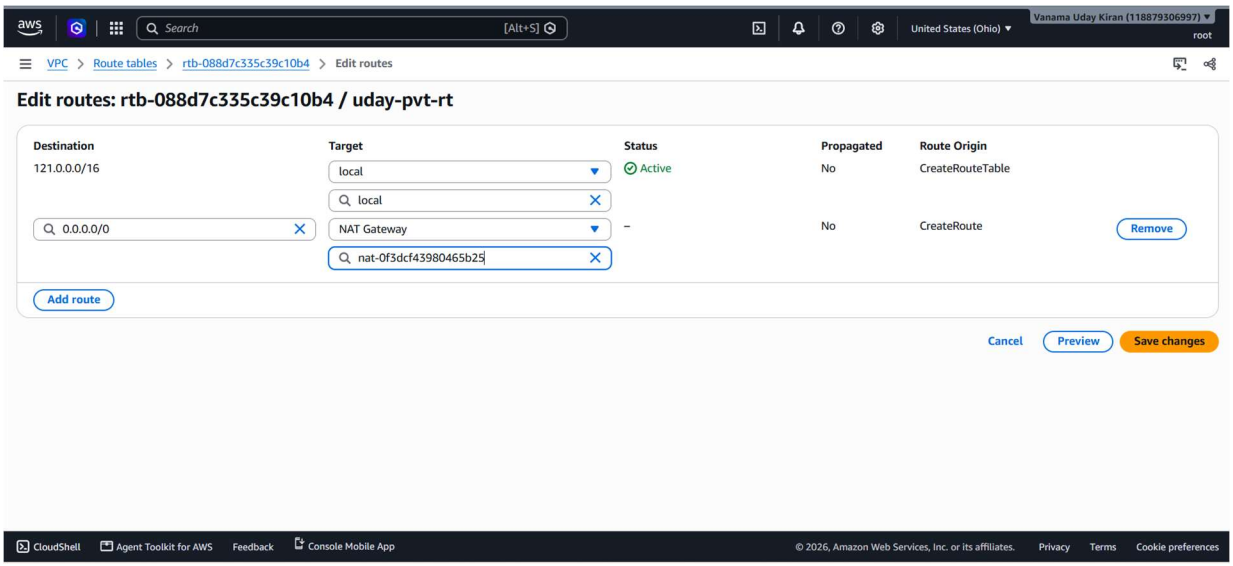


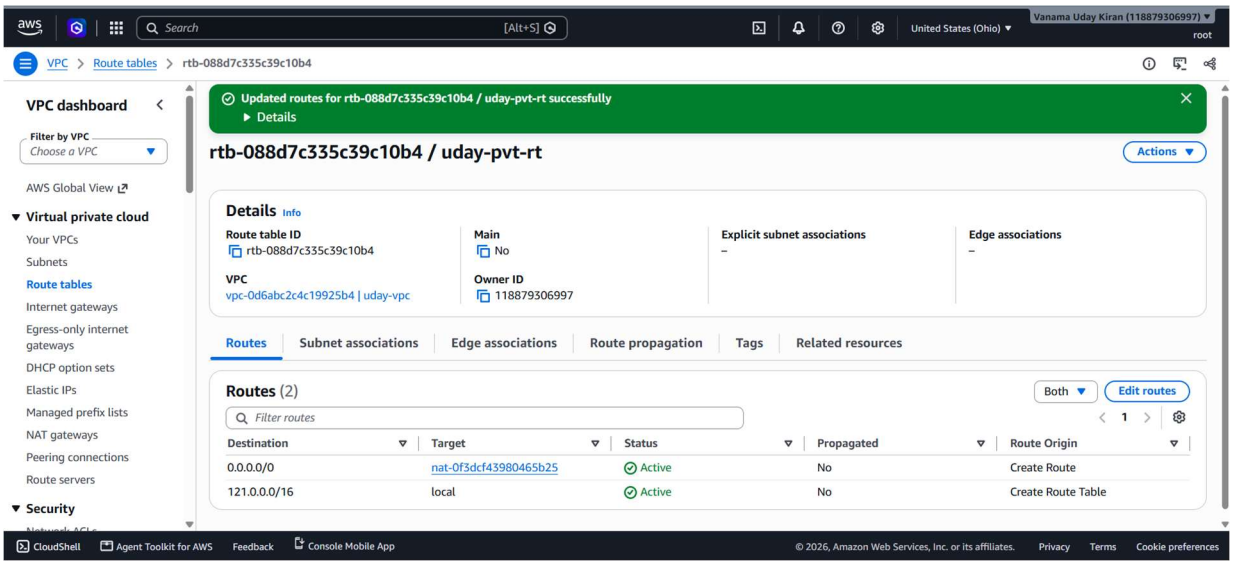




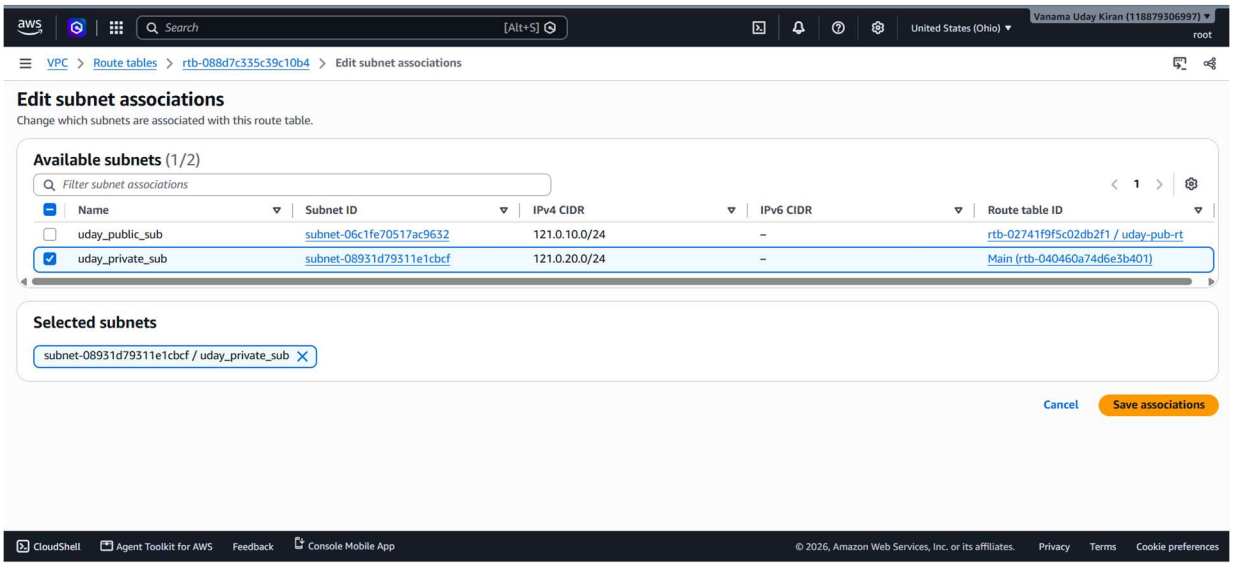
Connecting the NAT Gateway to the Private rt :

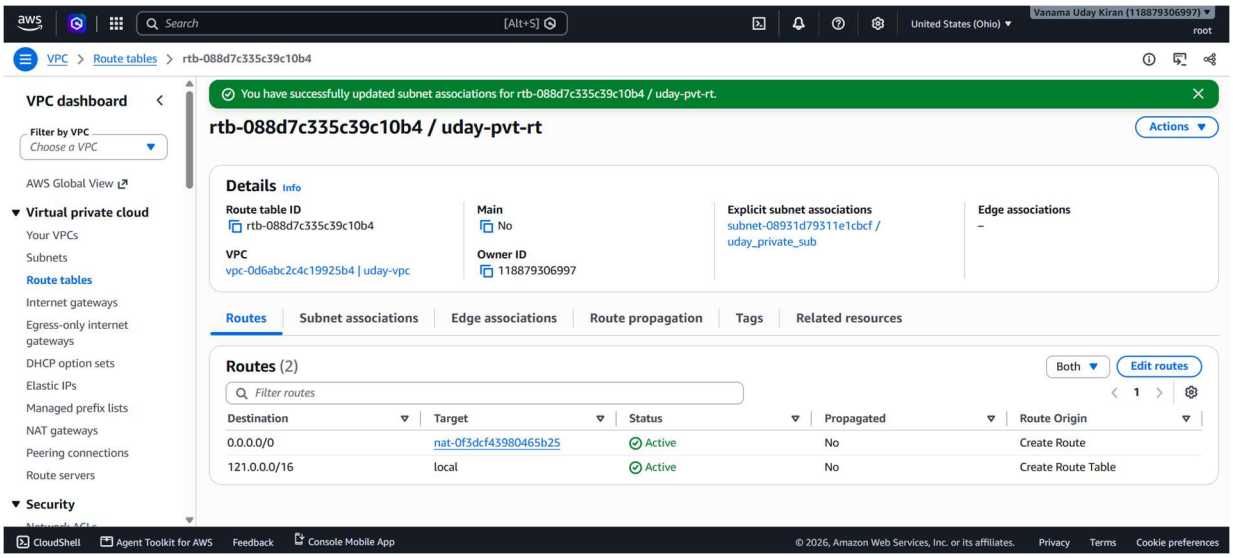




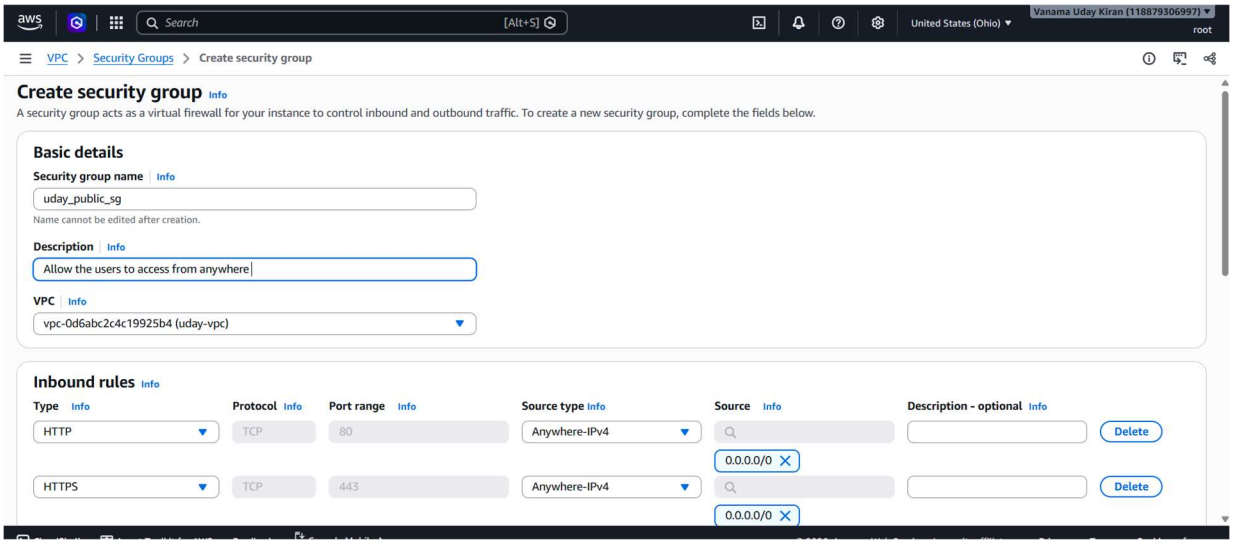


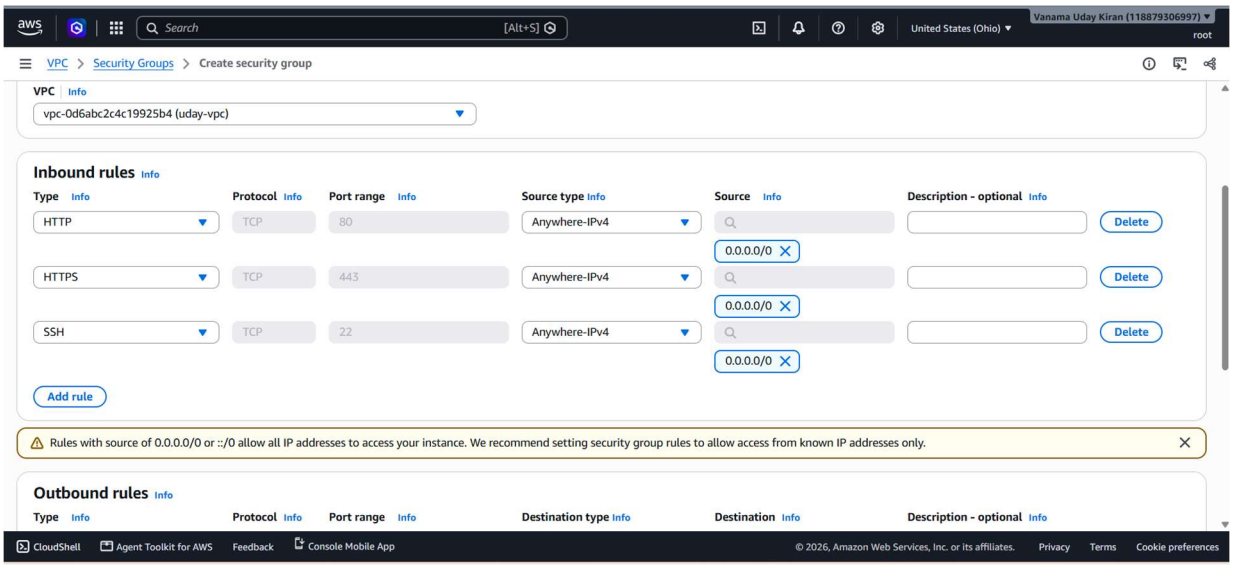
Edit The Subnet Associations :

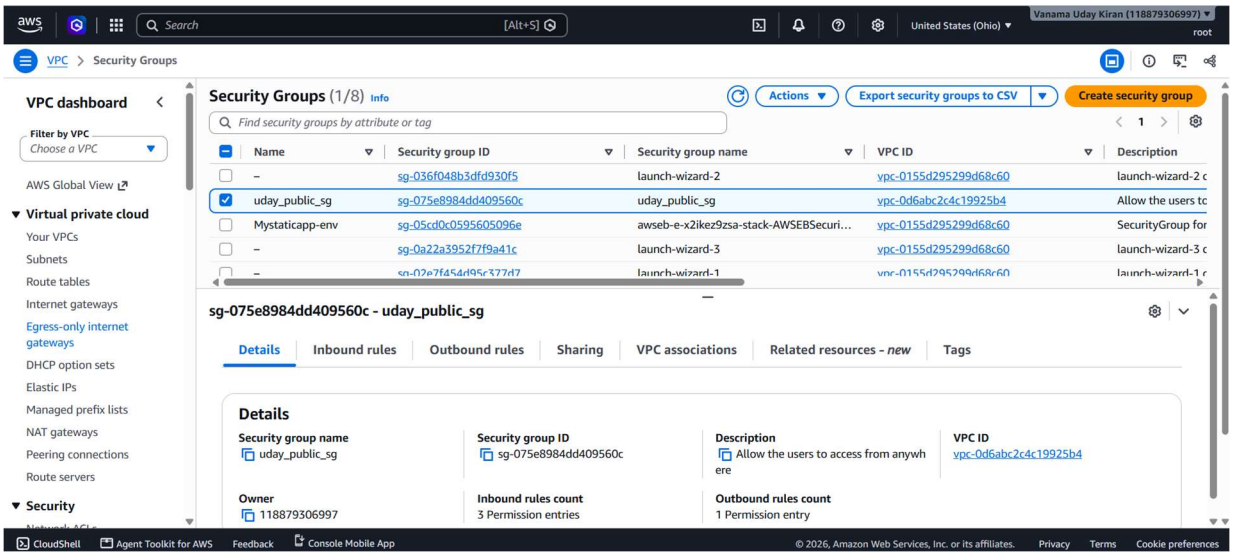


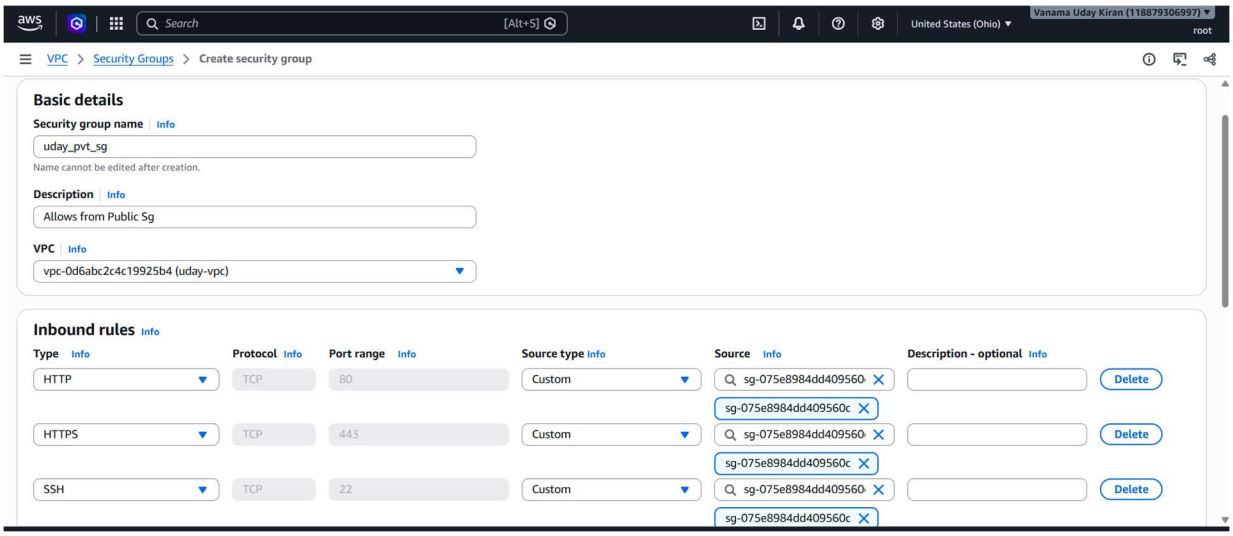


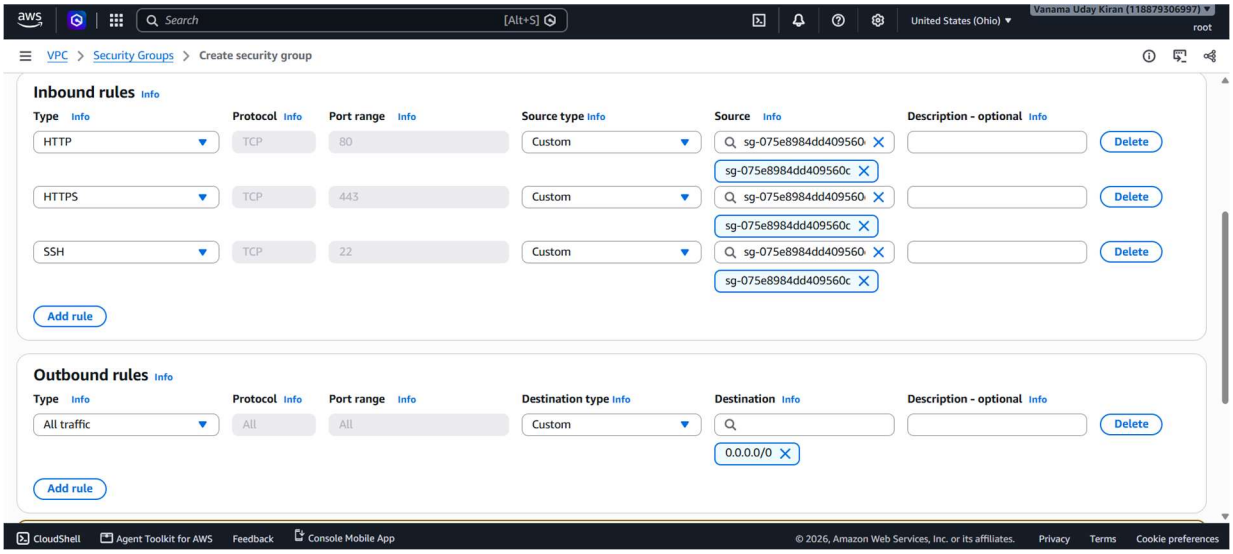
Creating The Security Groups :

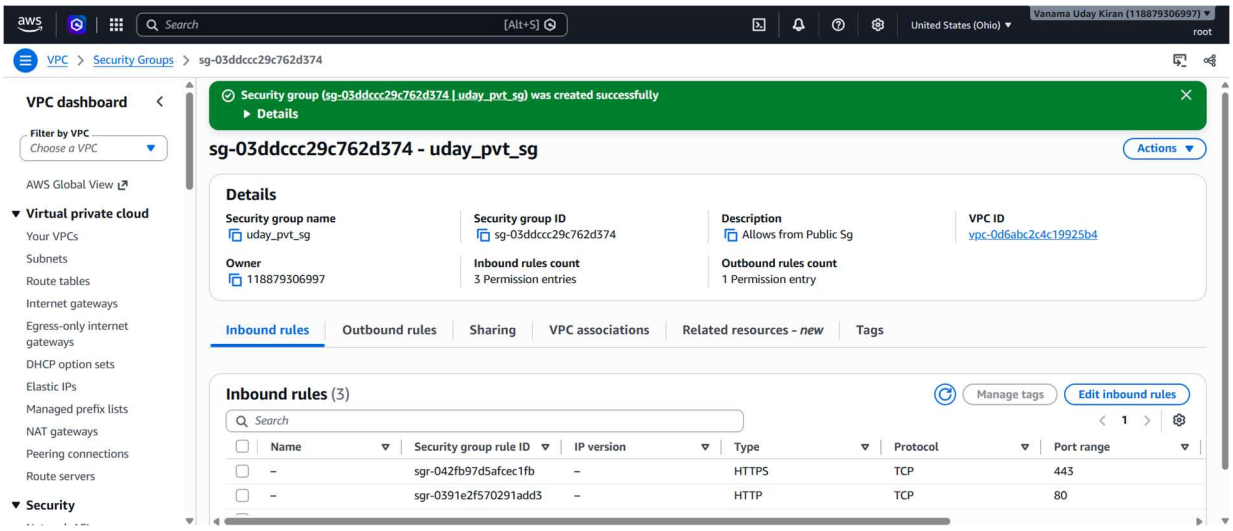


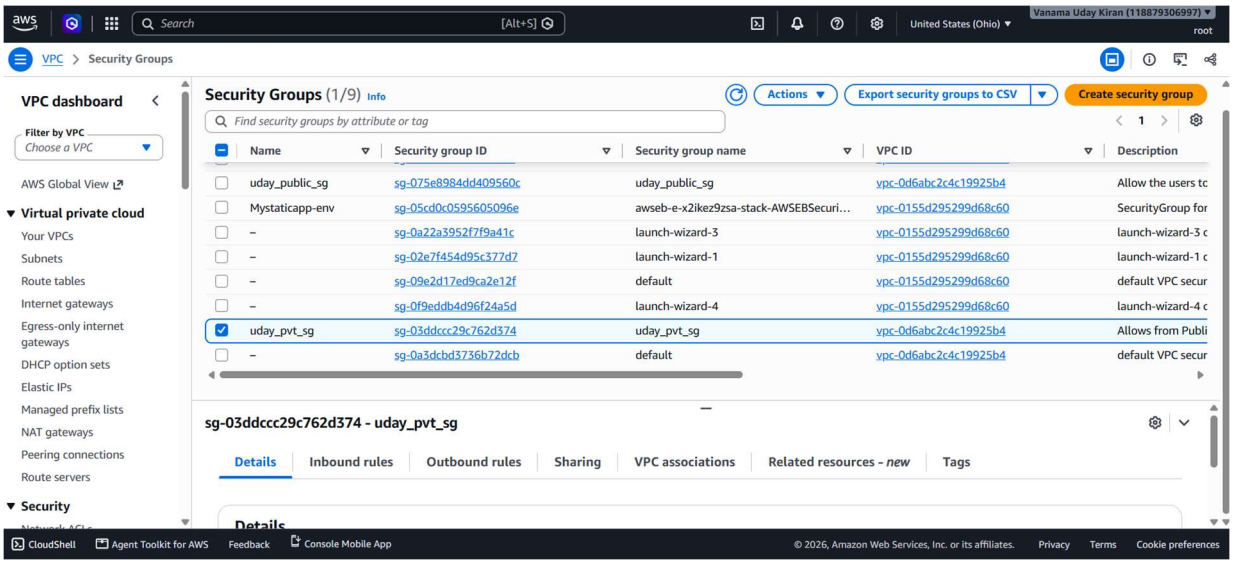

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United States (Ohio)
Vanama Uday Kiran (118879306997)
root
VPC > Security Groups > Create security group
VPC Info
vpc-0d6abc2c4c19925b4 (uday-vpc)
Inbound rules Info
Type Info
Protocol Info
Port range Info
Source type Info
Source Info
Description - optional Info
HTTP
TCP
80
Anywhere-IPv4
0.0.0.0/0
Delete
HTTPS
TCP
443
Anywhere-IPv4
0.0.0.0/0
Delete
SSH
TCP
22
Anywhere-IPv4
0.0.0.0/0
Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Outbound rules Info
Type Info
Protocol Info
Port range Info
Destination type Info
Destination Info
Description - optional Info
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root
VPC > Security Groups
VPC dashboard
Filter by VPC
Choose a VPC
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Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Security Groups (1/8) Info
Actions
Export security groups to CSV
Create security group
Find security groups by attribute or tag
Name
Security group ID
Security group name
VPC ID
Description
– sg-036f048b3dfd930f5 launch-wizard-2 vpc-0155d295299d68c60 launch-wizard-2 c
uday_public_sg sg-075e8984dd409560c uday_public_sg vpc-0d6abc2c4c19925b4 Allow the users to
Mystaticapp-env sg-05cd0c0595605096e awseb-e-x2ikez9zsa-stack-AWSEBSecuri… vpc-0155d295299d68c60 SecurityGroup for
– sg-0a22a3952f7f9a41c launch-wizard-3 vpc-0155d295299d68c60 launch-wizard-3 c
sg-075e8984dd409560c - uday_public_sg
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Details
Security group name
uday_public_sg
Security group ID
sg-075e8984dd409560c
Description
Allow the users to access from anywhere
VPC ID
vpc-0d6abc2c4c19925b4
Owner
118879306997
Inbound rules count
3 Permission entries
Outbound rules count
1 Permission entry
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United States (Ohio)
Vanama Uday Kiran (118879306997)
root
VPC > Security Groups > Create security group
Basic details
Security group name | Info
uday_pvt_sg
Name cannot be edited after creation.
Description | Info
Allows from Public Sg
VPC | Info
vpc-0d6abc2c4c19925b4 (uday-vpc)
Inbound rules Info
Type Info
Protocol Info
Port range Info
Source type Info
Source Info
Description - optional Info
HTTP
TCP
80
Custom
sg-075e8984dd409560c
Delete
HTTPS
TCP
443
Custom
sg-075e8984dd409560c
Delete
SSH
TCP
22
Custom
sg-075e8984dd409560c
Delete

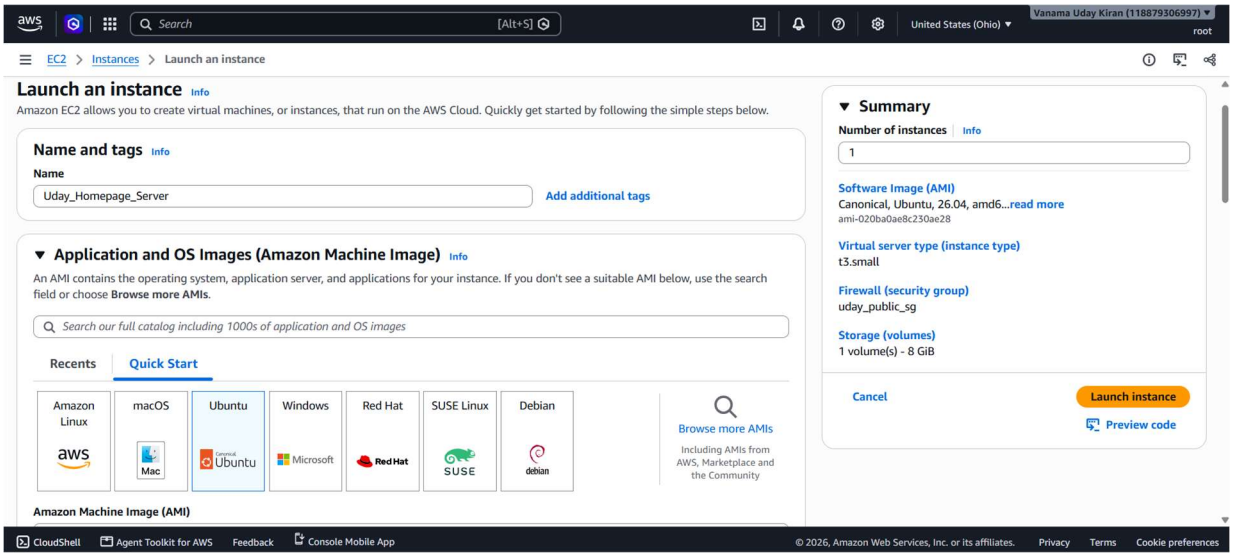

Search
[Alt+S]
United States (Ohio)
Vanama Uday Kiran (118879306997)
root
VPC > Security Groups > Create security group
Inbound rules Info
Type Info
Protocol Info
Port range Info
Source type Info
Source Info
Description - optional Info
HTTP
TCP
80
Custom
sg-075e8984dd409560c
Delete
HTTPS
TCP
443
Custom
sg-075e8984dd409560c
Delete
SSH
TCP
22
Custom
sg-075e8984dd409560c
Delete
Add rule
Outbound rules Info
Type Info
Protocol Info
Port range Info
Destination type Info
Destination Info
Description - optional Info
All traffic
All
All
Custom
0.0.0.0/0
Delete
Add rule
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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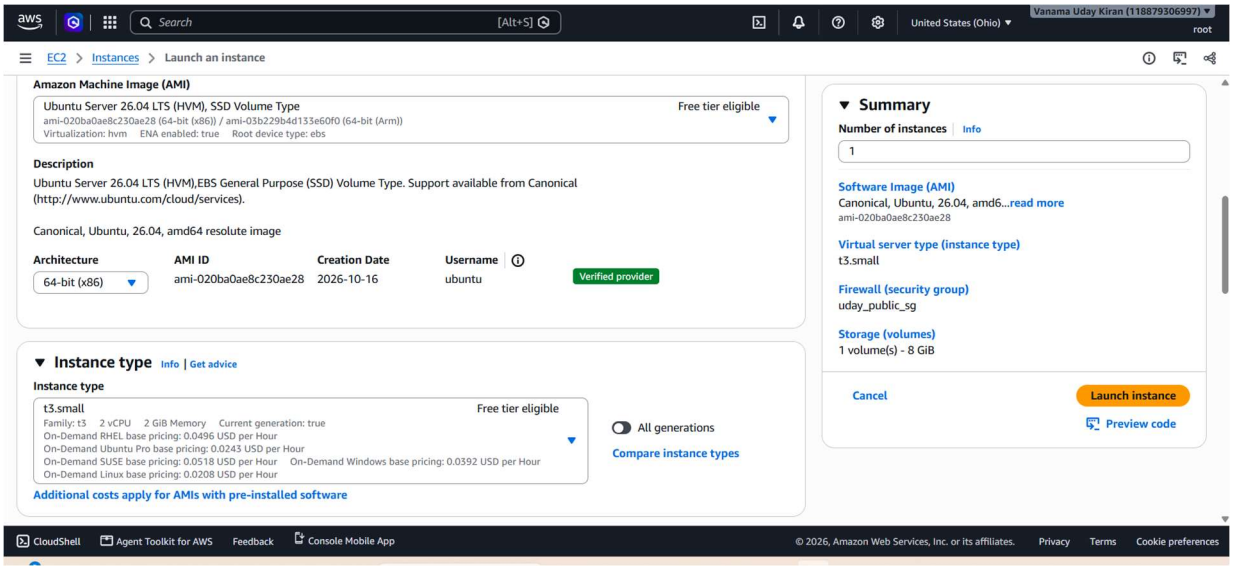

Search [Alt+S]
United States (Ohio)
Vanama Uday Kiran (118879306997)
root
VPC > Security Groups > sg-03ddccc29c762d374
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Security group (sg-03ddccc29c762d374 | uday_pvt_sg) was created successfully
Details
sg-03ddccc29c762d374 - uday_pvt_sg
Actions
Details
Security group name
uday_pvt_sg
Security group ID
sg-03ddccc29c762d374
Description
Allows from Public Sg
VPC ID
vpc-0d6abc2c4c19925b4
Owner
118879306997
Inbound rules count
3 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (3)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
– sgr-042fb97d5afcec1fb – HTTPS TCP 443
– sgr-0391e2f570291add3 – HTTP TCP 80

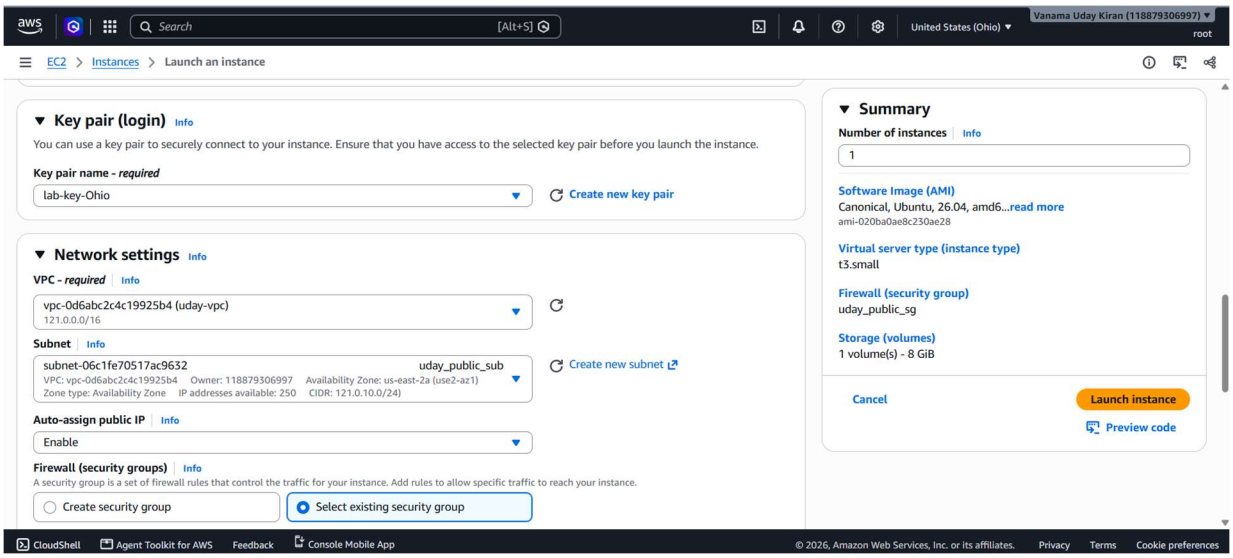

Search [Alt+S]
United States (Ohio)
Vanama Uday Kiran (118879306997)
root
VPC > Security Groups
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Security Groups (1/9) Info
Actions
Export security groups to CSV
Create security group
Find security groups by attribute or tag
Name
Security group ID
Security group name
VPC ID
Description
uday_public_sg sg-075e8984dd409560c uday_public_sg vpc-0d6abc2c4c19925b4 Allow the users tc
Mystaticapp-env sg-05cd0c0595605096e awseb-e-x2ikez9zsa-stack-AWSEBSecuri... vpc-0155d295299d68c60 SecurityGroup for
– sg-0a22a3952f7f9a41c launch-wizard-3 vpc-0155d295299d68c60 launch-wizard-3 c
– sg-02e7f454d95c377d7 launch-wizard-1 vpc-0155d295299d68c60 launch-wizard-1 c
– sg-09e2d17ed9ca2e12f default vpc-0155d295299d68c60 default VPC secur
– sg-0f9eddb4d96f24a5d launch-wizard-4 vpc-0155d295299d68c60 launch-wizard-4 c
uday_pvt_sg sg-03ddccc29c762d374 uday_pvt_sg vpc-0d6abc2c4c19925b4 Allows from Publi
– sg-0a3dcbd3736b72dcb default vpc-0d6abc2c4c19925b4 default VPC secur
sg-03ddccc29c762d374 - uday_pvt_sg
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Details
CloudShell
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Feedback
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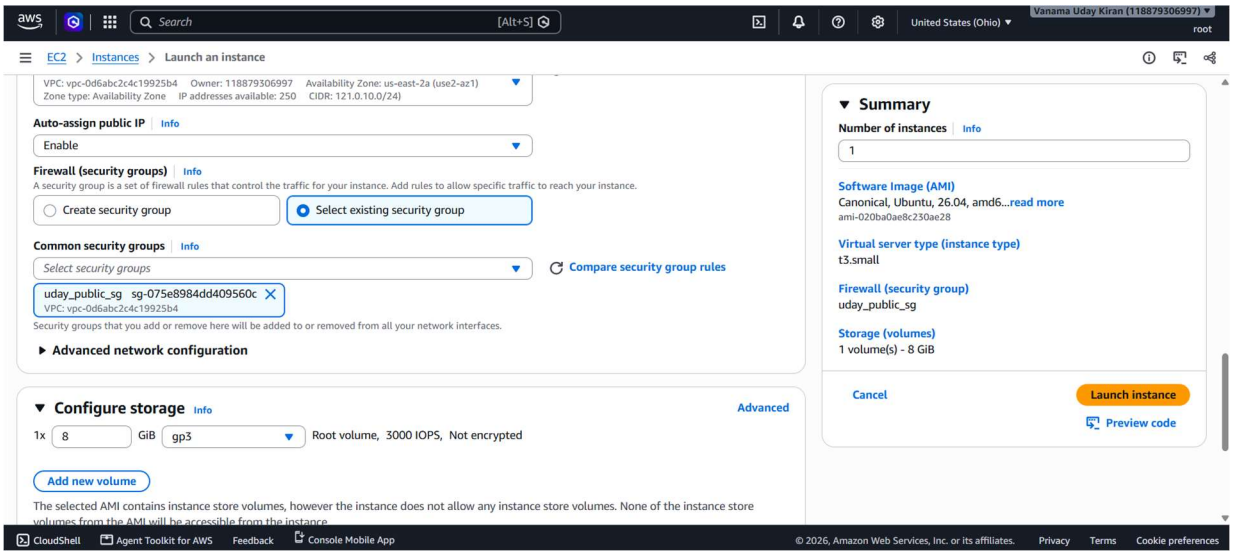
Creating 2 Ubuntu Instances:

1.Public Machine (Home page)

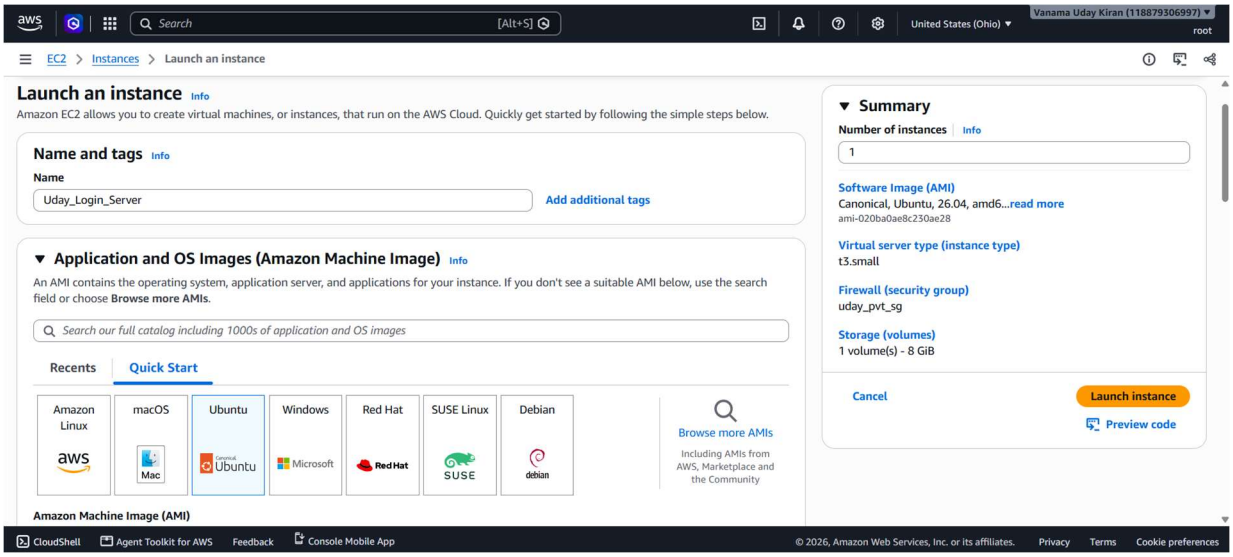


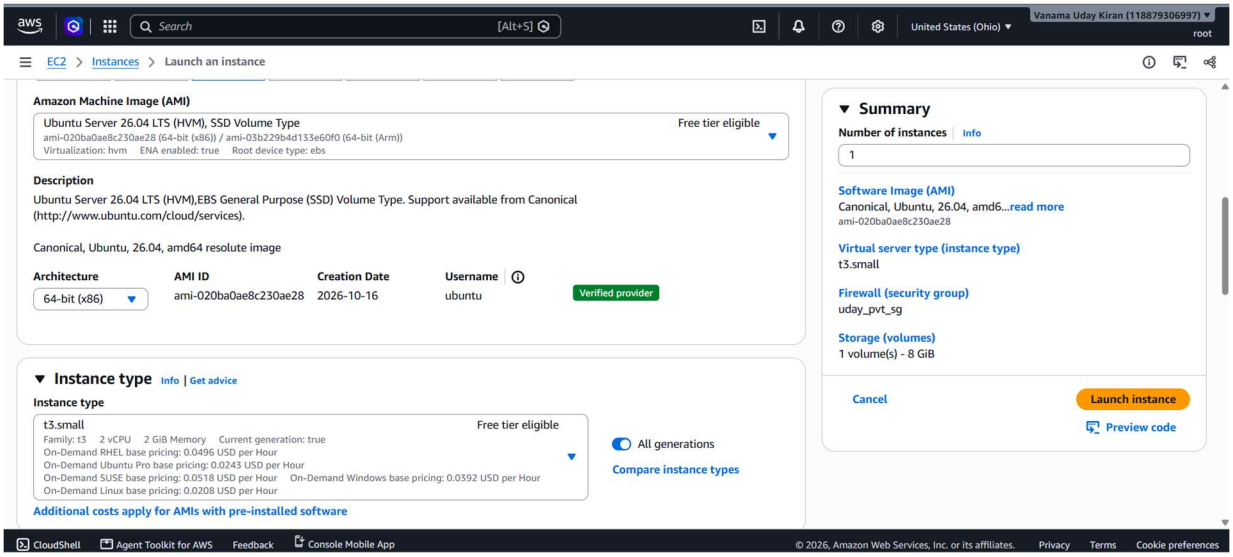


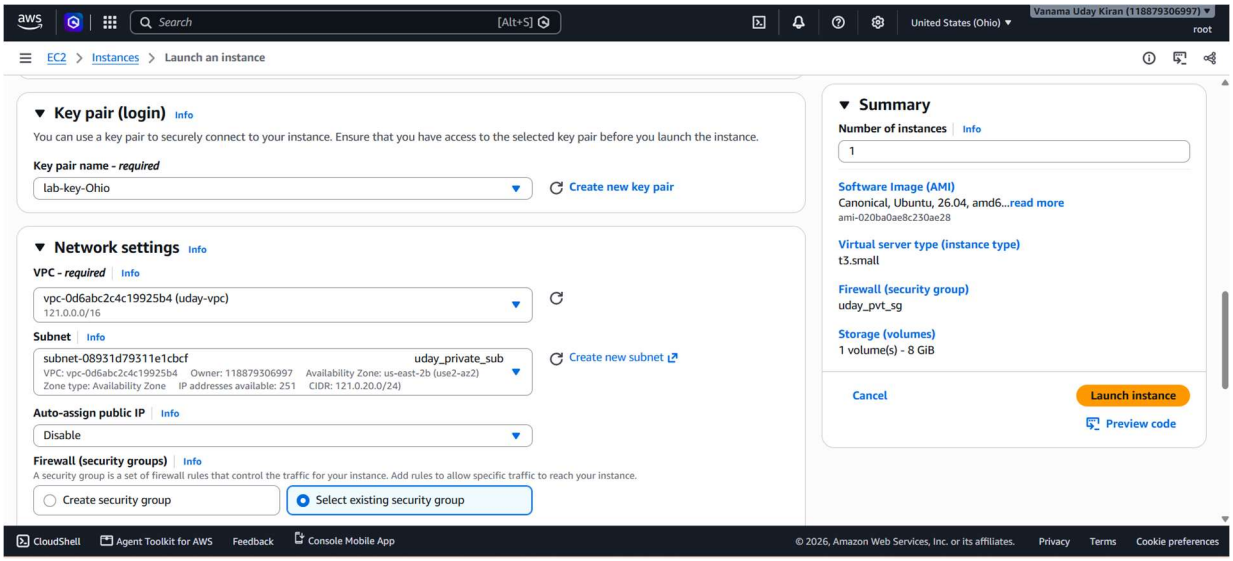

Search [Alt+S]
United States (Ohio)
Vanama Uday Kiran (118879306997)
root
EC2 > Instances > Launch an instance
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
lab-key-Ohio
Create new key pair
Network settings Info
VPC - required Info
vpc-0d6abc2c4c19925b4 (uday-vpc)
121.0.0.0/16
Subnet Info
subnet-06c1fe70517ac9632 uday_public_sub
VPC: vpc-0d6abc2c4c19925b4 Owner: 118879306997 Availability Zone: us-east-2a (use2-az1)
Zone type: Availability Zone IP addresses available: 250 CIDR: 121.0.10.0/24)
Create new subnet
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
ami-020ba0ae8c230ae28
Virtual server type (instance type)
t3.small
Firewall (security group)
uday_public_sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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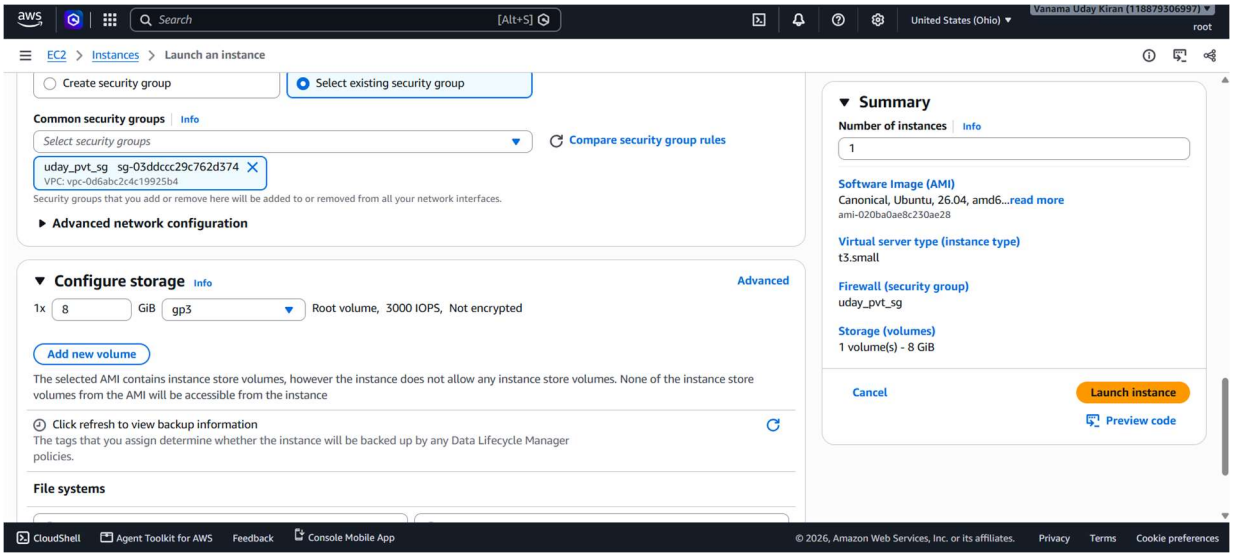

Search [Alt+S]
United States (Ohio)
Vanama Uday Kiran (118879306997)
root
EC2 > Instances > Launch an instance
VPC: vpc-0d6abc2c4c19925b4 Owner: 118879306997 Availability Zone: us-east-2a (use2-az1)
Zone type: Availability Zone IP addresses available: 250 CIDR: 121.0.10.0/24)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
uday_public_sg sg-075e8984dd409560c
VPC: vpc-0d6abc2c4c19925b4
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
ami-020ba0ae8c230ae28
Virtual server type (instance type)
t3.small
Firewall (security group)
uday_public_sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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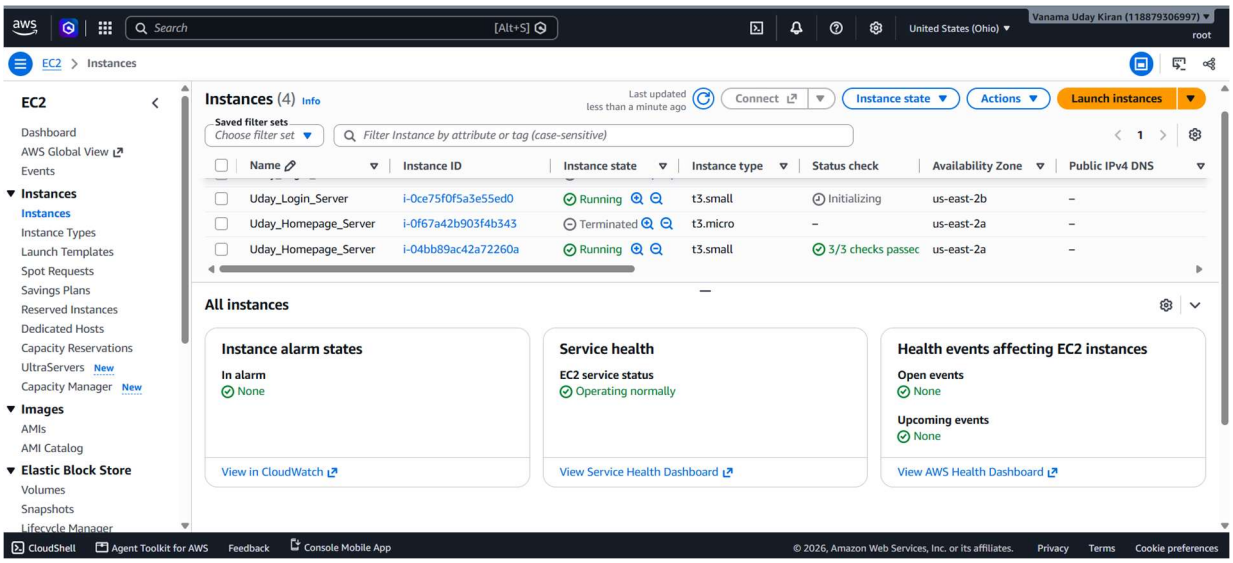
Creating the Login page Server:



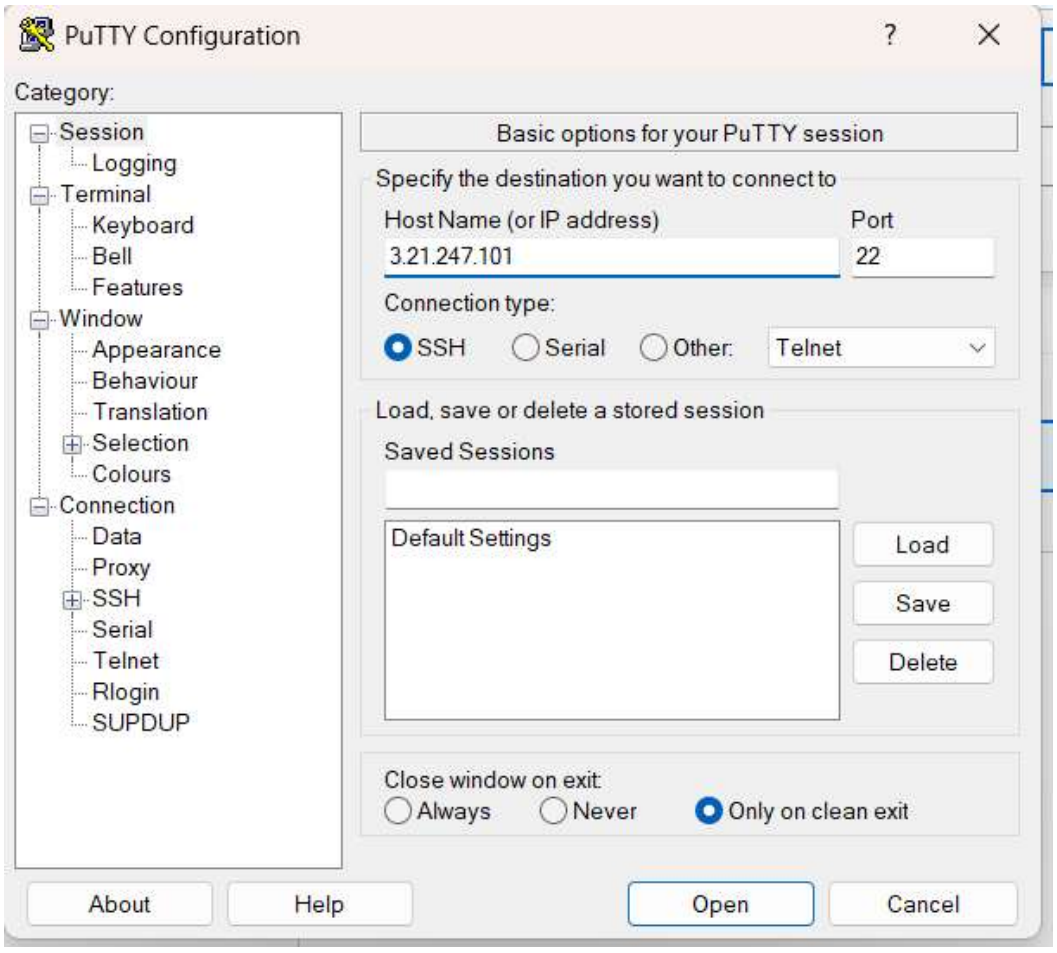



Search [Alt+S]
United States (Ohio)
Vanama Uday Kiran (118879306997)
root
EC2 > Instances > Launch an instance
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
lab-key-Ohio
Create new key pair
Network settings Info
VPC - required Info
vpc-0d6abc2c4c19925b4 (uday-vpc)
121.0.0.0/16
Subnet Info
subnet-08931d79311e1cbcf uday_private_sub
VPC: vpc-0d6abc2c4c19925b4 Owner: 118879306997 Availability Zone: us-east-2b (use2-az2)
Zone type: Availability Zone IP addresses available: 251 CIDR: 121.0.20.0/24)
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
ami-020ba0ae8c230ae28
Virtual server type (instance type)
t3.small
Firewall (security group)
uday_pvt_sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
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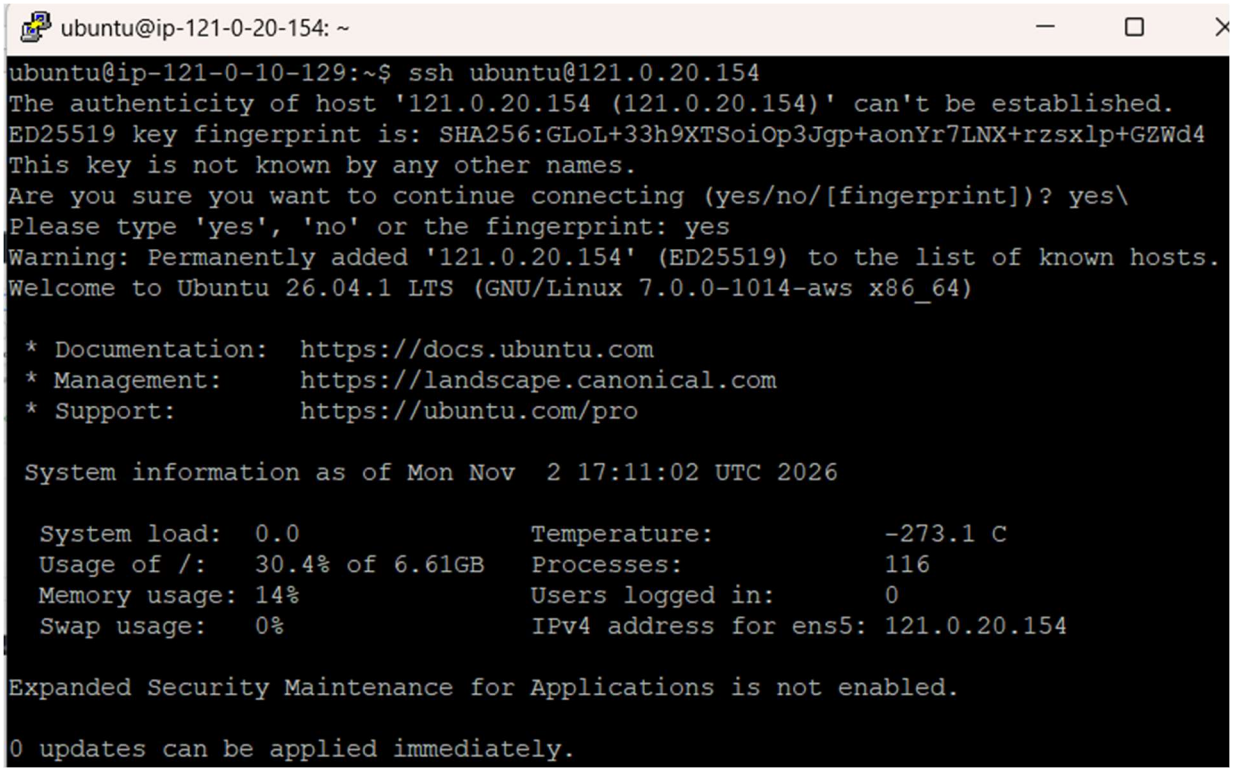

Search [Alt+S]
United States (Ohio)
Vanama Uday Kiran (118879306997)
root
EC2 > Instances > Launch an instance
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
uday_pvt_sg sg-03ddccc29c762d374
VPC: vpc-0d6abc2c4c19925b4
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
ami-020ba0ae8c230ae28
Virtual server type (instance type)
t3.small
Firewall (security group)
uday_pvt_sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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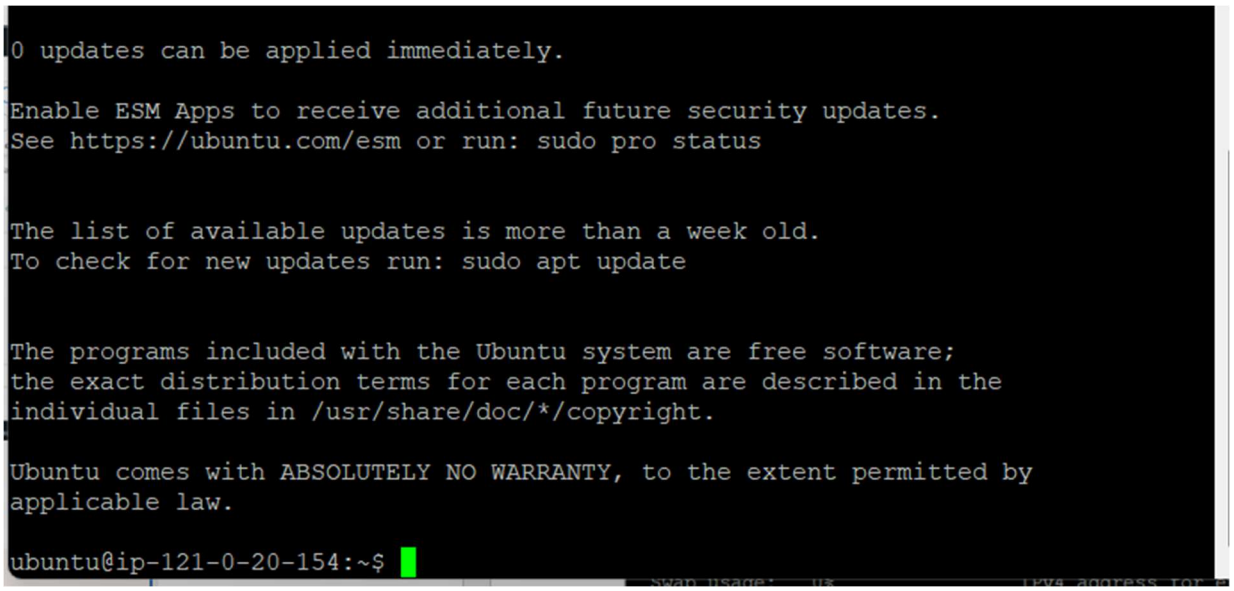


Connecting To the Public Server :



Connecting to The Private server Via Public Server :





Install Nginx:

```
ubuntu@ip-121-0-20-154:~$ sudo apt install -y nginx
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc  ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 31
  Download size: 667 kB
  Space needed: 1880 kB / 4680 MB available

Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64
v3 nginx-common all 1.28.3-2ubuntu1.12 [38.2 kB]
```

Creating the index.html Page in the Home page:

```
ubuntu@ip-121-0-20-154:~$ sudo tee /var/www/html/login/index.html << 'EOF'
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Login - Private Server</title>
    <style>
        body { font-family: Arial, sans-serif; display: flex; justify-content: c
enter; align-items: center; height: 100vh; margin: 0; background-color: #f0f2f5;
 }
        .login-card { background: #fff; padding: 30px; border-radius: 8px; box-s
hadow: 0 4px 12px rgba(0,0,0,0.1); width: 320px; text-align: center; }
        h2 { margin-top: 0; color: #333; }
        p { font-size: 0.85rem; color: #666; margin-bottom: 20px; }
        input { width: 100%; padding: 10px; margin: 8px 0; border: 1px solid #cc
c; border-radius: 4px; box-sizing: border-box; }
        button { width: 100%; padding: 10px; background-color: #28a745; color: w
hite; border: none; border-radius: 4px; font-size: 1rem; cursor: pointer; }
        button:hover { background-color: #218838; }
    </style>
</head>
<body>
    <div class="login-card">
```

Checking :

```
ubuntu@ip-121-0-20-154:~$ curl -I http://localhost/login/
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Mon, 02 Nov 2026 17:15:26 GMT
Content-Type: text/html
Content-Length: 1267
Last-Modified: Mon, 02 Nov 2026 17:14:28 GMT
Connection: keep-alive
ETag: "6ae8c4f4-4f3"
Accept-Ranges: bytes

ubuntu@ip-121-0-20-154:~$
```

Configure the Public Instance (3.21.247.101)

```
ubuntu@ip-121-0-10-129:~$ sudo tee /var/www/html/index.html << 'EOF'
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Home - Public Server</title>
    <style>
        body { font-family: Arial, sans-serif; text-align: center; padding-top:
100px; background-color: #fafafa; }
        h1 { color: #222; }
        p { color: #555; }
        .btn { display: inline-block; padding: 12px 24px; margin-top: 20px; back
ground-color: #007bff; color: white; text-decoration: none; border-radius: 5px;
font-weight: bold; font-size: 1.1rem; }
        .btn:hover { background-color: #0056b3; }
    </style>
</head>
<body>
    <h1>Welcome to the Application</h1>
    <p>This home screen is served from the <strong>Public Instance</strong>.</p>

    <a href="/login/" class="btn">Go to Login Page</a>
</body>
```

Configure NGINX Reverse Proxy:

Overwrite /etc/nginx/sites-available/default so requests to / serve the local home page, and requests to /login/ forward directly to the private IP:

ubuntu@ip-121-0-10-129: ~

```
server {
    listen 80 default_server;
    listen [::]:80 default_server;

    root /var/www/html;
    index index.html;

    server_name _;

    # Serve the local Home Page
    location / {
        try_files $uri $uri/ =404;
    }

    # Reverse Proxy for the Login Page on Private Instance
    location /login/ {
        proxy_pass http://121.0.20.154/login/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
EOF
server {
    listen 80 default_server;
    listen [::]:80 default_server;

    root /var/www/html;
    index index.html;

    server_name _;

    # Serve the local Home Page
    location / {
        try_files $uri $uri/ =404;
    }

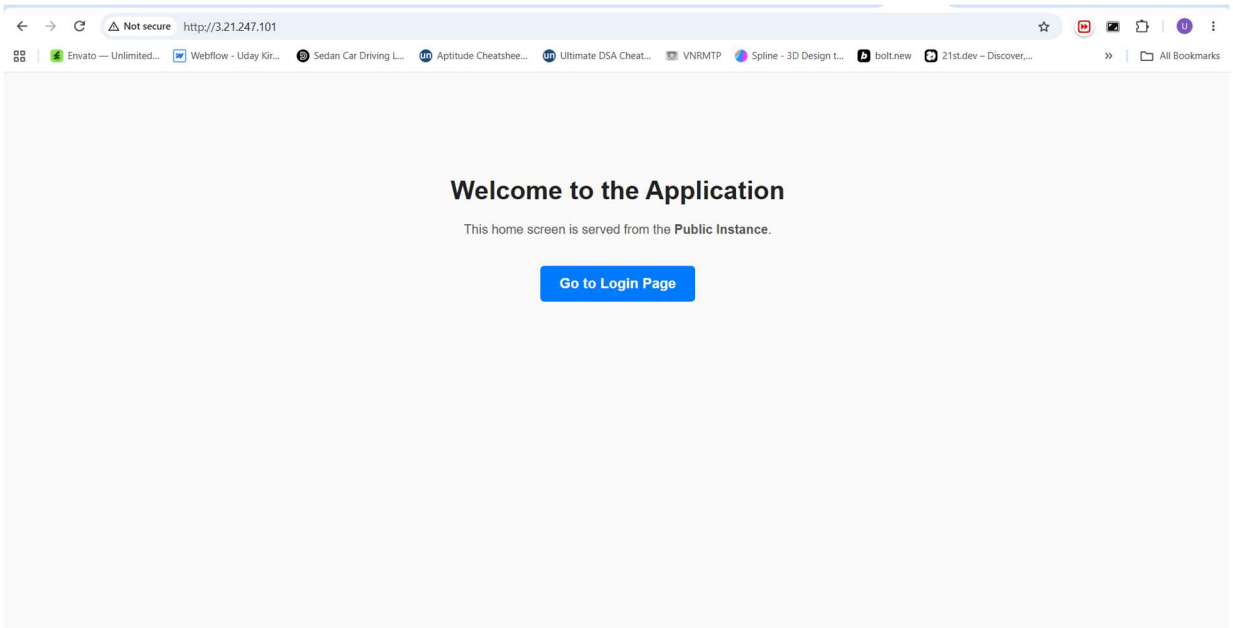
    # Reverse Proxy for the Login Page on Private Instance
    location /login/ {
        proxy_pass http://121.0.20.154/login/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
ubuntu@ip-121-0-10-129:~$
```

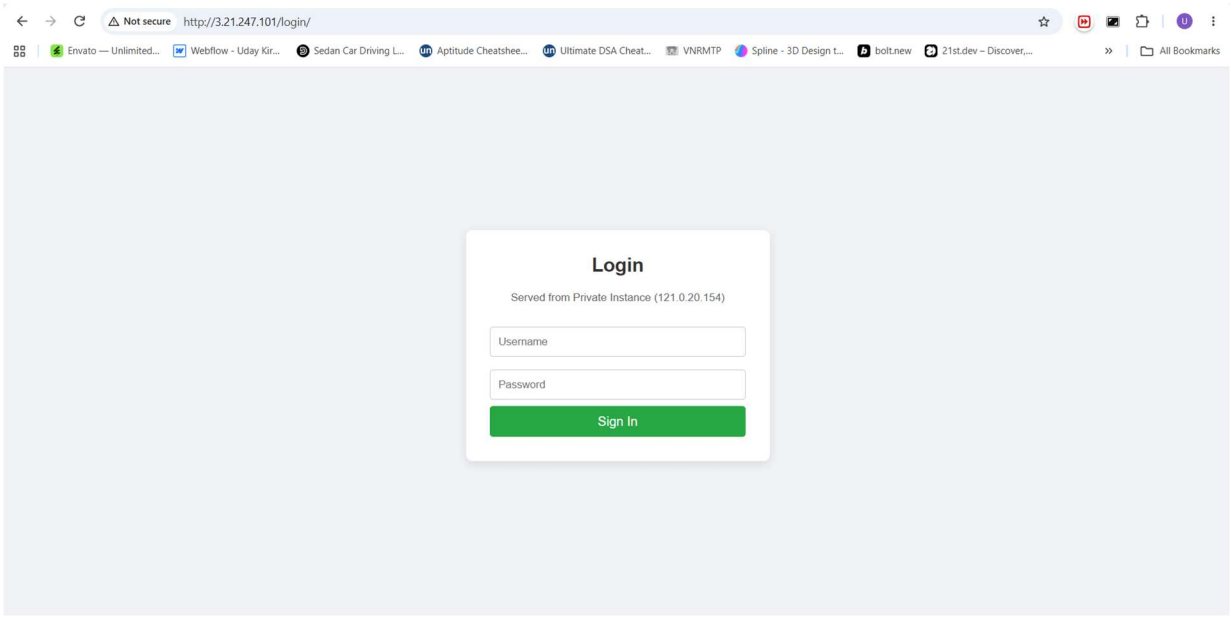
```
}
ubuntu@ip-121-0-10-129:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-121-0-10-129:~$ sudo systemctl restart nginx
ubuntu@ip-121-0-10-129:~$
```

Checking :

```
ubuntu@ip-121-0-10-129:~$ curl -I http://127.0.0.1/
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Mon, 02 Nov 2026 17:19:37 GMT
Content-Type: text/html
Content-Length: 760
Last-Modified: Mon, 02 Nov 2026 17:16:38 GMT
Connection: keep-alive
ETag: "6ae8c576-2f8"
Accept-Ranges: bytes

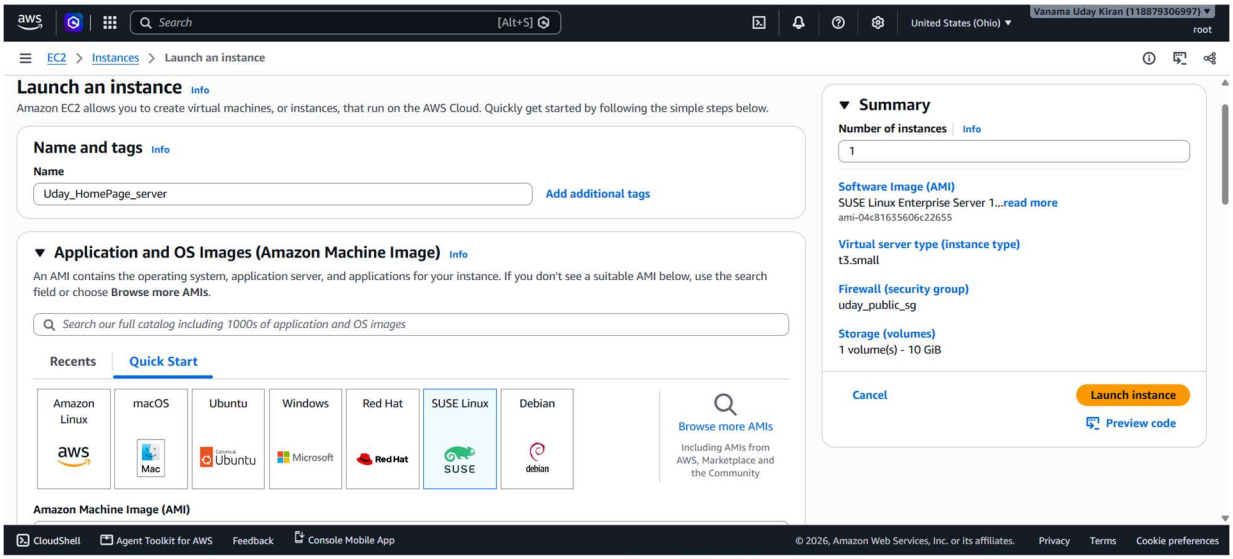
ubuntu@ip-121-0-10-129:~$ curl -I http://127.0.0.1/login/
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Mon, 02 Nov 2026 17:19:44 GMT
Content-Type: text/html
Content-Length: 1267
Connection: keep-alive
Last-Modified: Mon, 02 Nov 2026 17:14:28 GMT
ETag: "6ae8c4f4-4f3"
Accept-Ranges: bytes
```

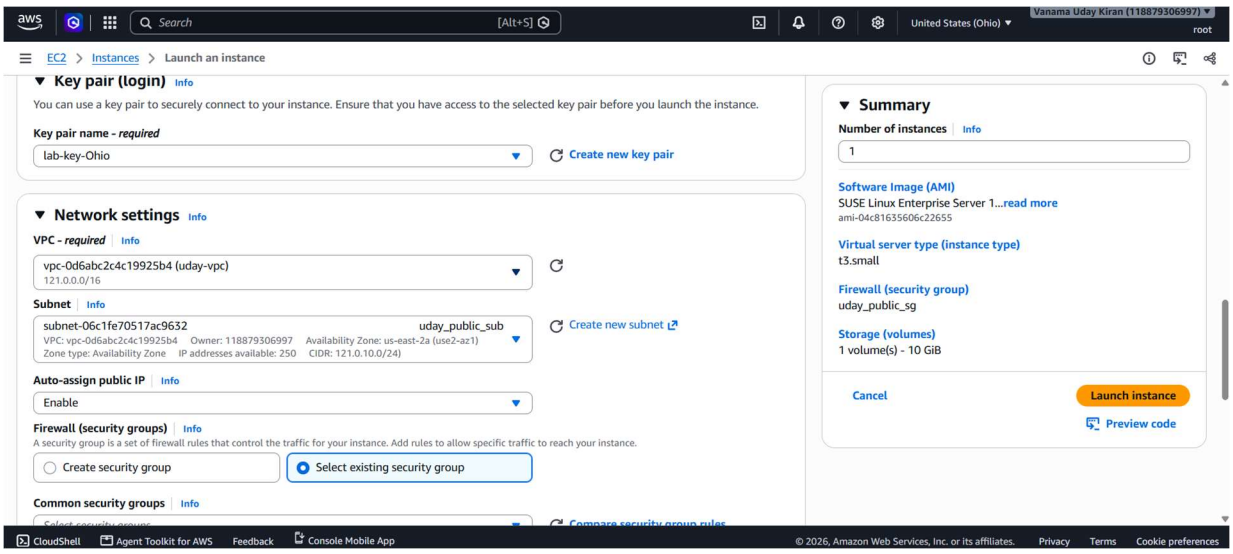


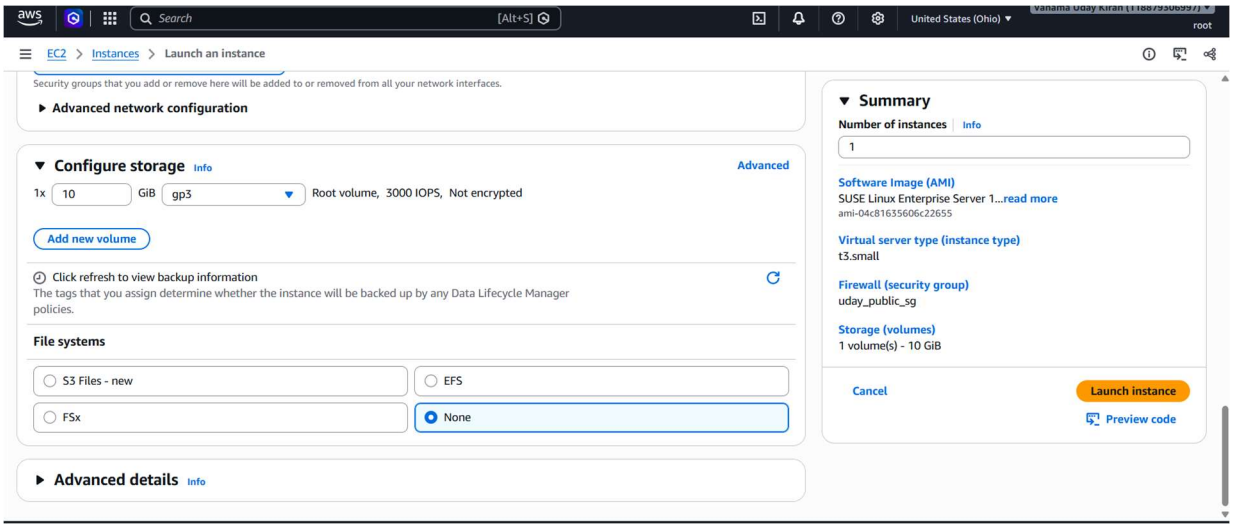

Not secure
http://3.21.247.101/login/
Envato — Unlimited...
Webflow - Uday Kir...
Sedan Car Driving L...
Aptitude Cheatshee...
Ultimate DSA Cheat...
VNRMTP
Spline - 3D Design t...
bolt.new
21st.dev – Discover,...
All Bookmarks
Login
Served from Private Instance (121.0.20.154)
Username
Password
Sign In

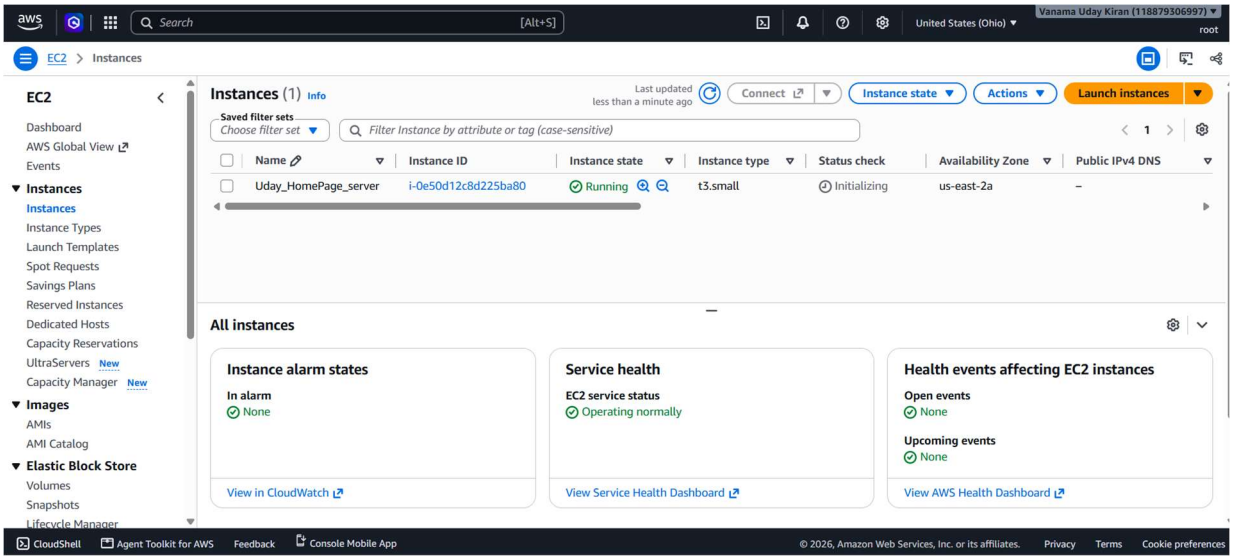
Testing With The SUSE Linux

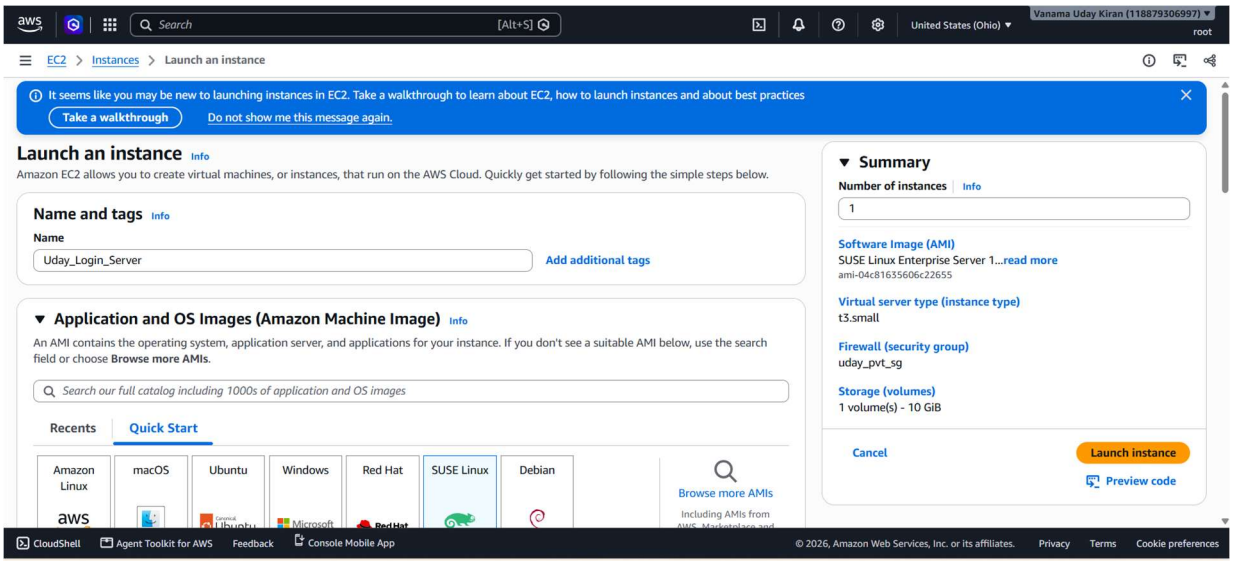
Creating The 2 Linux EC2 Instances Using the SUSI :

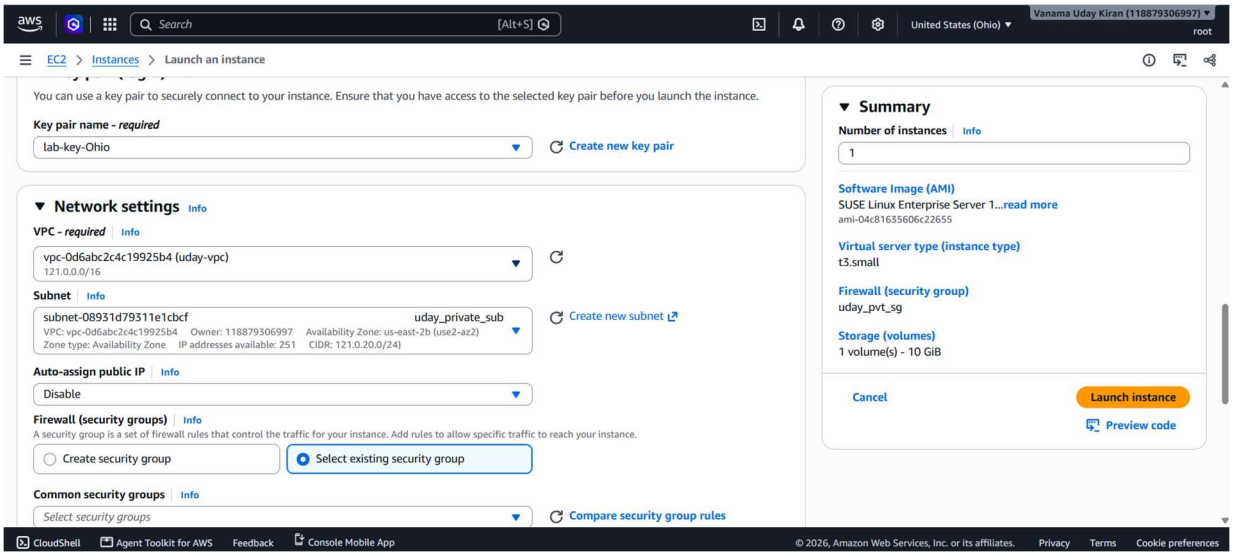


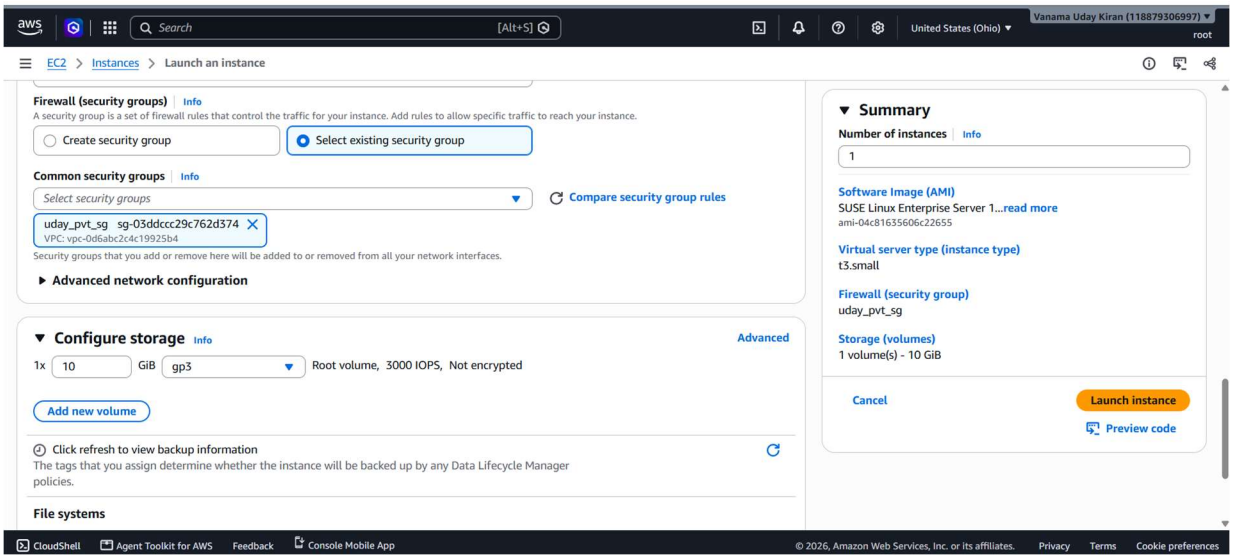


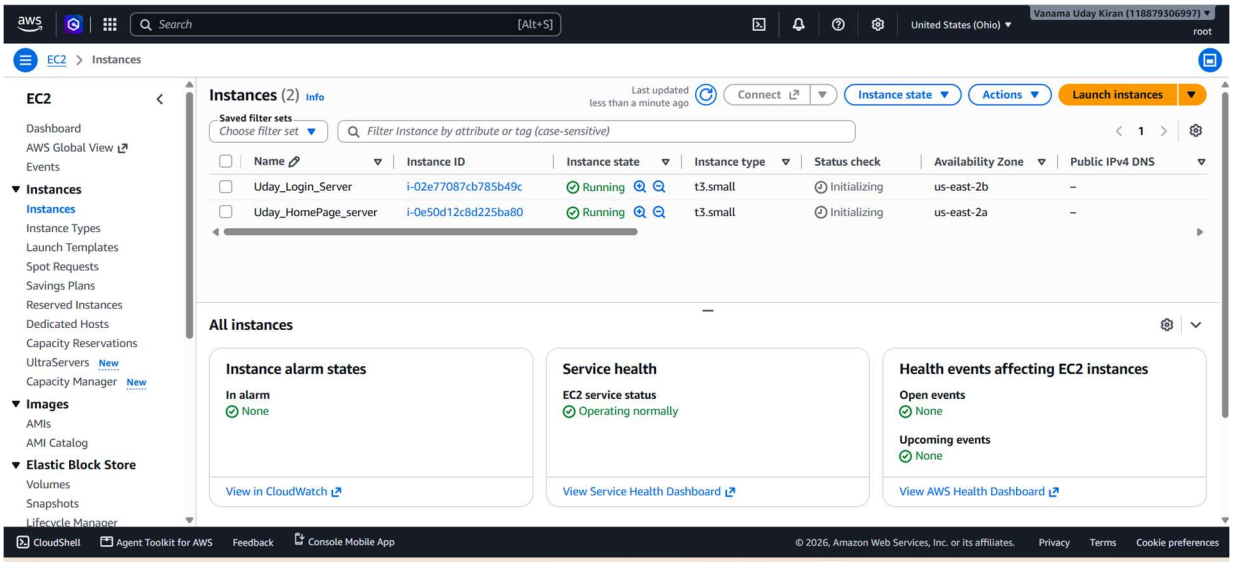

Search [Alt+S]
United States (Ohio)
root
EC2 > Instances > Launch an instance
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Configure storage Info
Advanced
1x 10 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Advanced details Info
Summary
Number of instances Info
1
Software Image (AMI)
SUSE Linux Enterprise Server 1...read more
ami-04c81635606c22655
Virtual server type (instance type)
t3.small
Firewall (security group)
uday_public_sg
Storage (volumes)
1 volume(s) - 10 GiB
Cancel
Launch instance
Preview code


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United States (Ohio)
Vanama Uday Kiran (118879306997)
root
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instances (1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Uday_HomePage_server
i-0e50d12c8d225ba80
Running
t3.small
Initializing
us-east-2a
–
All instances
Instance alarm states
In alarm
None
View in CloudWatch
Service health
EC2 service status
Operating normally
View Service Health Dashboard
Health events affecting EC2 instances
Open events
None
Upcoming events
None
View AWS Health Dashboard
CloudShell
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Feedback
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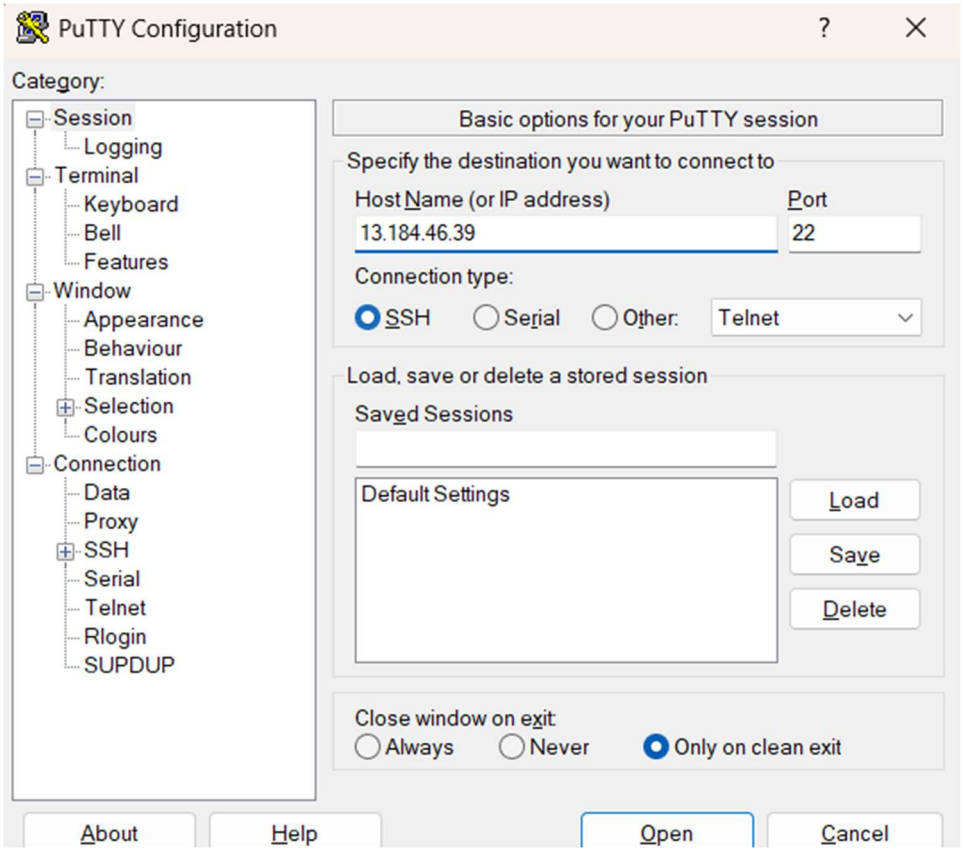

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United States (Ohio)
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root
EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
Uday_Login_Server
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from
Summary
Number of instances Info
1
Software Image (AMI)
SUSE Linux Enterprise Server 1...read more
ami-04c81635606c22655
Virtual server type (instance type)
t3.small
Firewall (security group)
uday_pvt_sg
Storage (volumes)
1 volume(s) - 10 GiB
Cancel
Launch instance
Preview code
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Agent Toolkit for AWS
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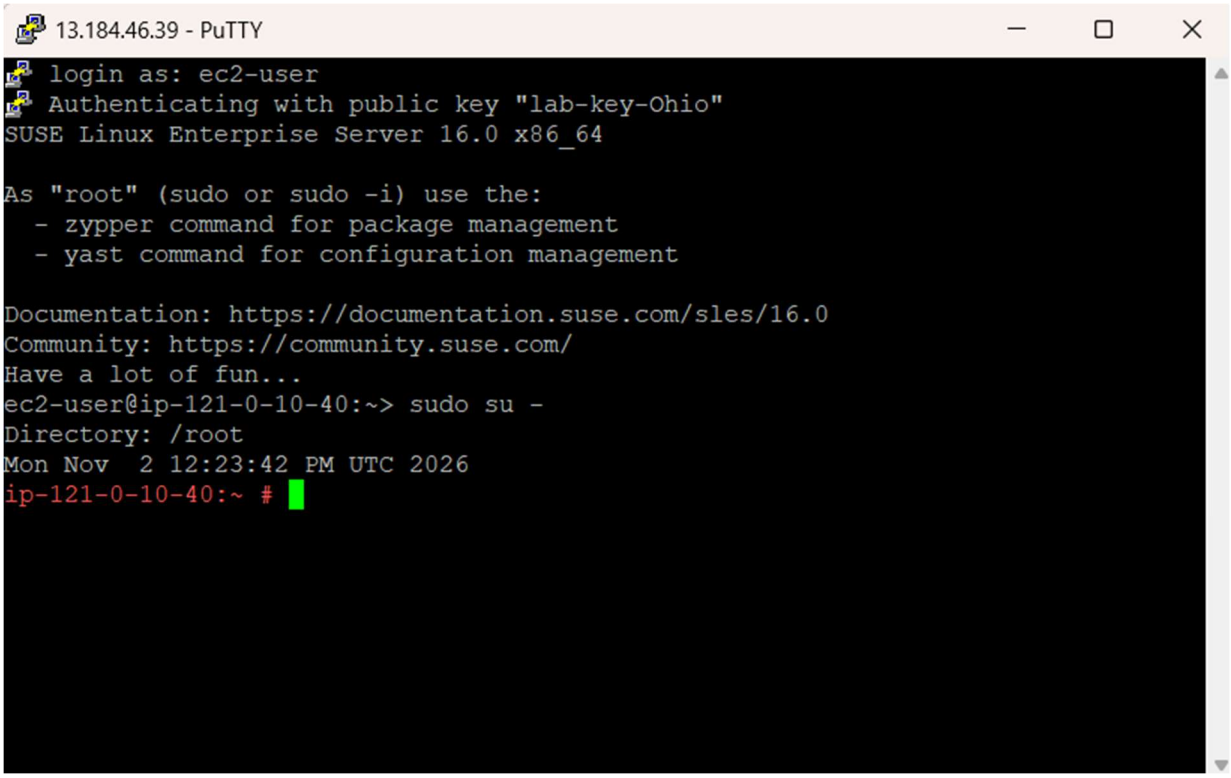

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root
EC2 > Instances > Launch an instance
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
lab-key-Ohio
Create new key pair
Network settings Info
VPC - required Info
vpc-0d6abc2c4c19925b4 (uday-vpc)
121.0.0.0/16
Subnet Info
subnet-08931d79311e1cbcf uday_private_sub
VPC: vpc-0d6abc2c4c19925b4 Owner: 118879306997 Availability Zone: us-east-2b (use2-az2)
Zone type: Availability Zone IP addresses available: 251 CIDR: 121.0.20.0/24)
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
Summary
Number of instances Info
1
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SUSE Linux Enterprise Server 1...read more
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t3.small
Firewall (security group)
uday_pvt_sg
Storage (volumes)
1 volume(s) - 10 GiB
Cancel
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Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
uday_pvt_sg sg-03ddccc29c762d374
VPC: vpc-0d6abc2c4c19925b4
Security groups that you add or remove here will be added to or removed from all your network interfaces.
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Configure storage Info
Advanced
1x 10 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
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Summary
Number of instances Info
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Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instances (2) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Uday_Login_Server
i-02e77087cb785b49c
Running
t3.small
Initializing
us-east-2b
–
Uday_HomePage_server
i-0e50d12c8d225ba80
Running
t3.small
Initializing
us-east-2a
–
All instances
Instance alarm states
In alarm
None
View in CloudWatch
Service health
EC2 service status
Operating normally
View Service Health Dashboard
Health events affecting EC2 instances
Open events
None
Upcoming events
None
View AWS Health Dashboard
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Connecting the Public home page Server:





Installing the apache Server:

```
ip-121-0-10-40:~ # sudo zypper install -y apache2
Refreshing service 'SUSE_Linux_Enterprise_Server_x86_64'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 2 recommended packages were automatically selected:
  apache2-utils w3m

The following 9 NEW packages are going to be installed:
  apache2 apache2-prefork apache2-utils libapr1-0 libapr-util1-0 libbrotlienc1
  libgc1 system-user-wwwrun w3m
```

```
ip-121-0-10-40:~ # sudo systemctl enable --now apache2
Created symlink '/etc/systemd/system/httpd.service' → '/usr/lib/systemd/system/a
pache2.service'.
Created symlink '/etc/systemd/system/apache.service' → '/usr/lib/systemd/system/
apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' →
'/usr/lib/systemd/system/apache2.service'.
ip-121-0-10-40:~ # sudo systemctl status apache2
● apache2.service - The Apache Webserver
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
     Active: active (running) since Mon 2026-11-02 12:25:57 UTC; 6s ago
 Invocation: a8550425f5eb4472a13dd81087ef9a0e
   Main PID: 2991 (httpd-prefork)
     Status: "Processing requests..."
      Tasks: 6
        CPU: 149ms
```

Changing the Permissions to edit the Html page :

```
ip-121-0-10-40:~ # ls -l /srv/www/htdocs/
total 0
ip-121-0-10-40:~ # sudo chown -R root:root /srv/www/htdocs/
ip-121-0-10-40:~ # sudo chmod -R 755 /srv/www/htdocs/
ip-121-0-10-40:~ #
```

```
ip-121-0-10-40:~ # sudo vi /srv/www/htdocs/index.html
            padding: 15px 38px;
            background: #00d4aa;
            color: #102b3c;
            text-decoration: none;
            font-weight: bold;
            border-radius: 8px;
            transition: 0.3s;
        }

        .login-btn:hover {
            background: #ffffff;
            transform: translateY(-3px);
        }

        .footer {
            margin-top: 35px;
            font-size: 12px;
            color: #b9d2e6;
        }
    </style>
</head>

<body>

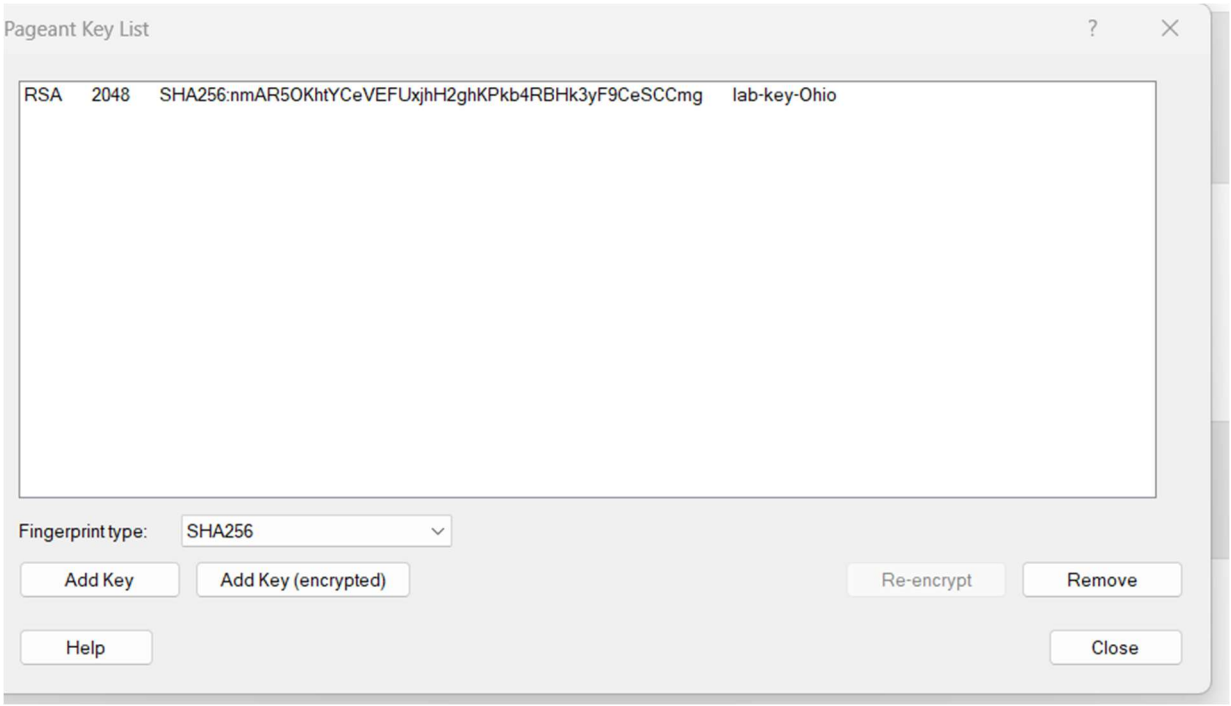
    <div class="container">
```

Given the private ip For the Login Button url:

```
    </p>

    <a href="http://121.0.20.209/login.html" class="login-btn">
        Go to Login
```

Adding the Key to Pagent :



Connect to the Private Server :

```
ec2-user@ip-121-0-10-40:~> ssh ec2-user@121.0.20.209
The authenticity of host '121.0.20.209 (121.0.20.209)' can't be established
ED25519 key fingerprint is SHA256:vkrskaVF1YoiIX6565HiF0Egc9PAvAgU5OXCs9S+g
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '121.0.20.209' (ED25519) to the list of known ho
SUSE Linux Enterprise Server 16.0 x86_64

As "root" (sudo or sudo -i) use the:
  - zypper command for package management
  - yast command for configuration management

Documentation: https://documentation.suse.com/sles/16.0
Community: https://community.suse.com/
Have a lot of fun...
ec2-user@ip-121-0-20-209:~>
```

Installing Apache :

```
ec2-user@ip-121-0-20-209:~> sudo zypper install -y apache2
Refreshing service 'SUSE_Linux_Enterprise_Server_x86_64'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 2 recommended packages were automatically selected:
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The following 9 NEW packages are going to be installed:
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  libgc1 system-user-wwwrun w3m

9 new packages to install.

Package download size:     4.9 MiB

Package install size change:
              |       9.9 MiB  required by packages that will be installed
    9.9 MiB   |  -      0 B    released by packages that will be removed

Backend:  classic_rpmtrans
Continue? [y/n/v/...? shows all options] (y): y
Preloading: libapr1-0-1.7.5-160000.2.2.x86_64.rpm [done]
Preloading: libbrotlienc1-1.1.0-160000.2.2.x86_64.rpm [done]
Preloading: libgc1-8.2.8-160000.2.2.x86_64.rpm [done]
Preloading: system-user-wwwrun-20170617-160000.2.2.noarch.rpm [done]
Preloading: libapr-util1-0-1.6.3-160000.3.1.x86_64.rpm [done]
Preloading: w3m-0.5.3+git20230121-160000.2.2.x86_64.rpm [done]
Preloading: apache2-utils-2.4.66-160000.4.1.x86_64.rpm [done]
Preloading: apache2-prefork-2.4.66-160000.4.1.x86_64.rpm [done]
Preloading: apache2-2.4.66-160000.4.1.x86_64.rpm [done]
```

Enabling :

```
Running post-transaction scripts .....................................[done]
ec2-user@ip-121-0-20-209:~> sudo systemctl enable --now apache2
Created symlink '/etc/systemd/system/httpd.service' → '/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/apache.service' → '/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.
rvice'.
```

```
ec2-user@ip-121-0-20-209:~> sudo chown -R root:root /srv/www/htdocs/
ec2-user@ip-121-0-20-209:~> sudo chmod -R 755 /srv/www/htdocs/
ec2-user@ip-121-0-20-209:~> sudo vi /srv/www/htdocs/login.html
ec2-user@ip-121-0-20-209:~>
```

13.184.46.39 - PuTTY

```
            background: #065c68;
        }

        .footer {
            text-align: center;
            margin-top: 22px;
            font-size: 12px;
            color: #888;
        }
    </style>
</head>

<body>

    <div class="login-box">

        <h2>Welcome Back</h2>

        <p class="subtitle">Sign in to your Cloud Login Portal</p>

        <form>
            <label>Username</label>
            <input type="text" placeholder="Enter username" required>

            <label>Password</label>
            <input type="password" placeholder="Enter password" required>

            <button type="submit">Login</button>
        </form>

        <div class="footer">
            Secure Private Network Access
        </div>

    </div>

</body>
</html>
"/srv/www/htdocs/login.html" 111L, 2407B
```

Making the Proxy Requests Off

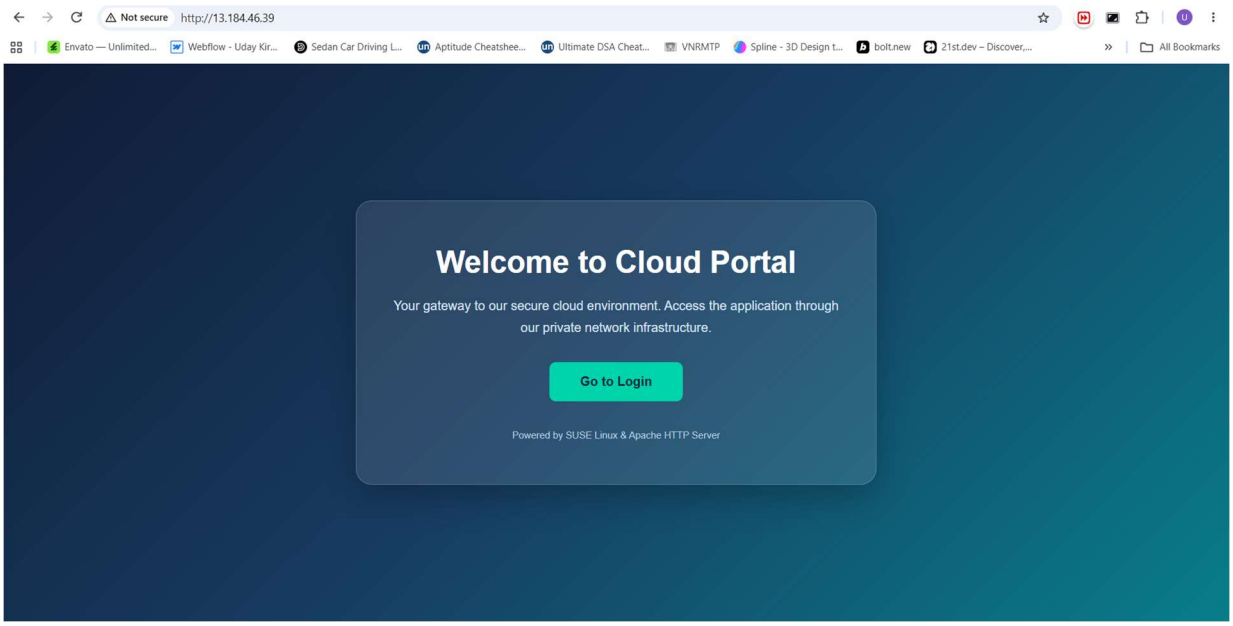
The proxy Requests will be changes That makes the redirection Not work directly so we need to change that

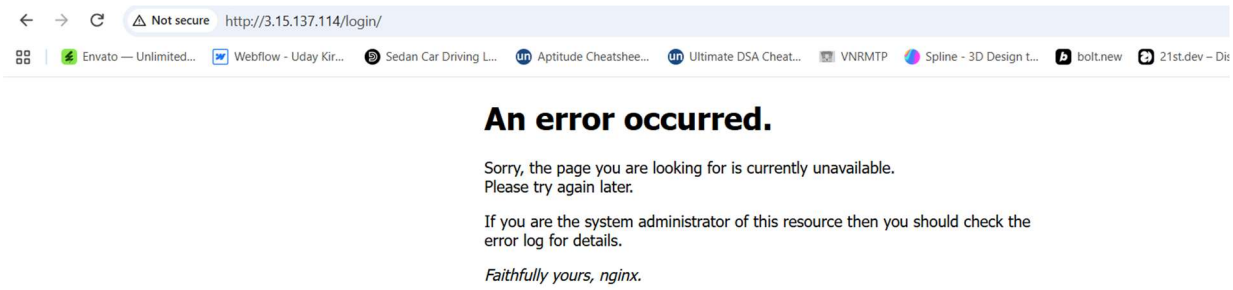
Configure Reverse Proxy on Public SUSI

```
Connection to 121.0.20.209 closed.
ec2-user@ip-121-0-10-40:~> sudo systemctl restart apache2
ec2-user@ip-121-0-10-40:~> sudo a2enmod proxy
sudo a2enmod proxy_httpec2-user@ip-121-0-10-40:~>
ec2-user@ip-121-0-10-40:~> sudo vi /etc/apache2/conf.d/login-proxy.conf
ProxyRequests Off

ProxyPass /login http://121.0.20.209/login.html
ProxyPassReverse /login http://121.0.20.209/login.html
~
~
~
~
~
~
~
~
~
~
~
~
~
```

Checking :





Got the Error Due to The ARMORS and Security Groups In SUSE Linux ……